
Unsupervised Wavetable Seam Artifact Repair via Hybrid GA–PPO Meta-Search

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Code (ReelSynth engine + DenoiseOpt meta-search): <https://github.com/reeldemo/reelsynth>,
<https://github.com/reeldemo/denoise-opt-meta>

ABSTRACT

In single-period wavetable synthesis, a sample discontinuity at the wrap boundary injects a broadband transient that repeats at the fundamental frequency, producing an audible periodic click. We address this *cycle-local seam artifact* with **DenoiseOpt**, a residual-scored hybrid meta-search that selects and fits bake-cell operators Θ applied at the period boundary. The bake cell composes classical DualCosine crossfades, FIR blends, and optional lightweight residual networks, possibly routed via Mixture-of-Experts (MoE) soft gating, without requiring paired studio-clean recordings. Candidates maximize the *prolonged residual score* $R \in [0, 1]$ against a procedural ideal sibling $r^* = G(\text{seed}, \text{cliff}=\text{off})$; the cracked engine is $x = G(\text{seed}, \text{cliff}=\text{on})$ from the same draw. No-bake leaves x unchanged and scores that unrepaired period against r^* ; it is not the ideal. DualCosine may center PPO advantages only; ranking still maximizes absolute R . The outer search loop interleaves genetic algorithm (GA) tournament selection, PPO-based architecture proposals, and optional population-based training (PBT). On a frozen canonical sine+cliff holdout (seed 20260719), the compact meta-favorite achieves $R \approx 0.991$ vs classical no-bake $R \approx 0.972$, *seam_fir3* $R \approx 0.961$, and DualCosine $R \approx 0.825$ (gaps vs classical board). Hard-cliff strata (top 10% wrap-jump, $n=410$) keep favorite $R \approx 0.9915$ while DualCosine falls to $R \approx 0.657$; no-bake R stays high (≈ 0.967) because residual mass is small relative to ideal RMS (passthrough $x \mapsto x$). A same-generator corrupt $\rightarrow$ corrupt Noise2Noise control reaches $R \approx 0.992$; a supervised LSTM sibling ceiling reaches $R \approx 0.995$. Wavetable-native transfer uses true ReelSynth-exported factory periods ($n=25$) and external AKWF CC0 single-cycle WAVs ($n=24$) under the same wrap protocol (no LibriSpeech/MUSDB). On exports the favorite trails no-bake / FIR (domain shift); on AKWF it matches no-bake and beats DualCosine. Scope is strictly cycle-local wavetable seam restoration (DAFx / AES / arXiv cs.SD); we do not claim speech enhancement or deep SOTA on sci/eng transfer boards.

Keywords unsupervised learning; wavetable; wrap discontinuity; seam restoration; DenoiseOpt; prolonged residual score; neural architecture search; reinforcement learning; genetic algorithms; meta-learning

1 Introduction

Wavetable oscillators store a single period of a waveform and advance a read pointer modulo the table length. When $x[0] \neq x[L-1]$, the wrap is a hard discontinuity that, under cyclic playback, becomes a periodic click with a broadband spectrum locked to the fundamental. The same geometry appears whenever a short loop buffer is tiled for listening or for objective evaluation. We define the repair task as *cycle-local seam restoration*: edit a neighborhood of the wrap (and optionally a lightweight residual cell over the period) so that multi-period playback matches an ideal reference that never opens the wrap cliff. That outer agreement is scored by the prolonged residual R for this periodic seam artifact class.

Figure 1 shows one real sine-wrap edge case from the frozen canonical holdout (same `make_batch` tile, seed 20260719). Four roles must stay distinct: (a) ideal sibling r^* (cliff withheld; unscored target); (b) no-bake / cracked

engine x after open-wrap cliff ($R \approx 0.950$ vs r^* ; ideal overlay); (c) DualCosine classical bake on that same tile ($R \approx 0.603$ on this severe cliff); (d) Ours / DenoiseOpt favorite bake (`champion_iter_000235`, tile $R \approx 0.9917$). Panels (e)–(f) plot the modulation residual and a seam zoom so Ours is readable against DualCosine. No-bake is not the ideal; DualCosine is one classical row (and separately a PPO advantage baseline; Section 3).

Classical virtual-analog practice already supplies seam-local tools. Alias-free BLIT/BLEP-style synthesis [1], PolyBLEP / fractional-delay antialiasing [3], and BLAMP corner corrections [2] reduce wrap and geometric discontinuity energy in-band. Those operators act locally; we score cycle-local bake approximations of them beside DualCosine (Section 5.7). They leave open which bake parameters and which residual cell, if any, minimize prolonged tiled error against a no-cliff ideal. Label-free restoration work such as

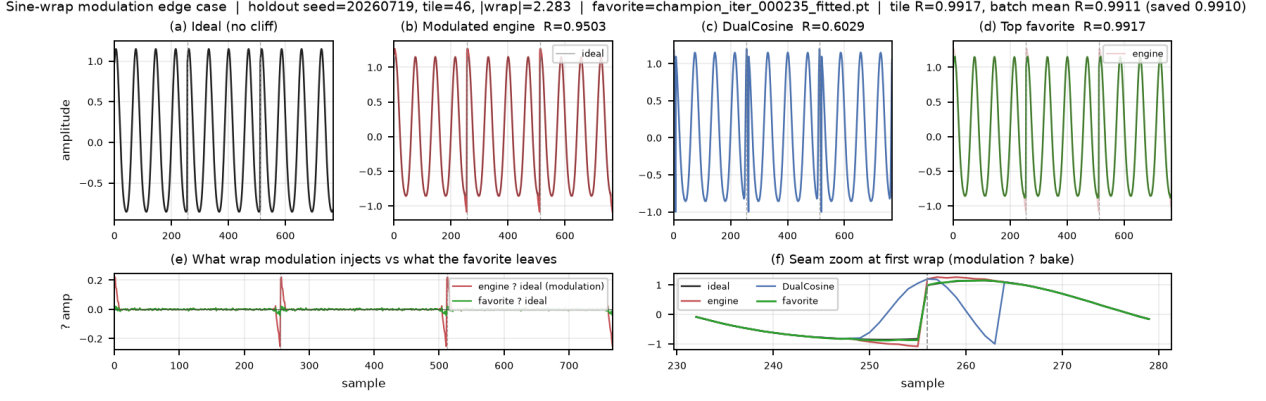


Figure 1: Canonical holdout sine-wrap edge case (seed 20260719, $L=256$, $N=16$ tiles, same tile across panels). Distinguish four roles: (a) ideal sibling r^* (cliff withheld; scoring target, not a method); (b) no-bake / cracked engine x (passthrough $\Theta=id$, scored vs r^*); (c) DualCosine classical end-fade bake; (d) Ours (DenoiseOpt favorite bake). (e)–(f) Modulation residual and seam zoom. Prolonged residual $R \in [0, 1]$ (1 = best) is printed per panel where scored. Accessibility: six-panel waveform comparison of ideal, no-bake, DualCosine, and Ours.

Noise2Noise shows that useful reconstructors can be ranked without paired clean targets when the evaluation geometry is explicit [4]. Speech-domain Noise2Noise denoisers extend that idea to audio waveforms [5]. Mixture-invariant unsupervised separation likewise avoids paired clean stems for ranking [29]. On the search side, NAS surveys [16], RL controllers [17, 18], aging evolution [20, 21], population-based hyperparameter schedules [22], CMA-ES tutorials [24], and evolution-guided policy gradients [23] motivate a hybrid outer loop: GA for population diversity, PPO for discrete architecture edits [19], and PBT-style exploit–mutate on elites. Platform-aware and progressive NAS add compute-aware search priors [27, 28]. MoE soft gates suggest routing among heterogeneous tiny cells under one residual score [25]. Waveform separation nets (Wave-U-Net, Conv-TasNet, dual-path RNN, SepFormer) and audio DL surveys inform the searchable cell family at bake-scale width [7–9, 11, 12, 30, 31], with U-Net / ResNet / attention primitives as block ancestors [10, 13–15].

The contribution is a meta-learning protocol for this periodic seam instance. An outer RL+GA(+PBT) loop proposes bake-cell operator graphs and continuous bake parameters. Each candidate is scored by prolonged residual $R \in [0, 1]$ (1 = best) between engine-baked multi-period audio and a procedural ideal sibling $r^* = G(\text{seed}, \text{cliff}=\text{off})$. Paired studio-clean wavetables are not required for every trial. Differentiable DSP modules [6] place DualCosine / FIR / MLP seam ops in an interpretable stack. Bayesian HPO [26] informs prior families that compete under the same R . A CUDA hybrid search campaign (PPO+GA+PBT+NAS, depth and MoE biases) has passed a 5,000-iteration clean gate with champion $R \approx 0.9909$ versus DualCosine ≈ 0.8166 on the runner geometry. The frozen canonical holdout and inference benches in Section 5 are the measured evaluation matrices for this report. Primary claim scope is cycle-local wavetable seam restoration (DAFx / AES / arXiv cs.SD). We do not claim speech en-

hancement, music source-separation, or deep SOTA on classical sci/eng transfer boards.

1.1 Contributions

1. **Problem framing.** Cast wavetable wrap discontinuity as cycle-local seam restoration under prolonged tiled evaluation, separating seam-local diagnostics from residual R .
2. **Hybrid DenoiseOpt outer loop.** Specify RL+GA(+PBT) search over discrete operator graphs and continuous bake parameters, with named branches (GA crossover, PPO mutation, PBT exploit, MoE soft gates).
3. **Formal residual protocol.** Define bake operator Θ , ideal-sibling generator G , outer objective $\max R(y, r^*)$, and reimplementable Algorithms 1–9 (Appendix A). No wrap-closure or search-convergence guarantees.
4. **Open evaluation path.** Release companion code for the synthesizer engine and meta search, with frozen 5k-gate figures, hard-cliff strata, N2N/seq baselines, and wavetable-native transfer benches.

Section 2 organizes related work by mechanism. Sections 3–4 detail the residual objective and search protocol. Section 5 reports the 5k-gate freeze and inference benches.

2 Related Work

We organize prior work by mechanism. **Used** lines are methodological or comparative anchors for residual-scored RL+GA meta-search. **Screened** lines were inspected as contrast. They are excluded from the dependency set. Paraphrases prefer attached abstracts and PDF snippets over title-only citation.

2.1 Discontinuity control and modular audio DSP

Alias-free digital synthesis of classic analog waveforms (BLIT/BLEP lineage) states the wrap discontinuity problem for trivial oscillators [1]. PolyBLEP-style fractional-delay corrections give practical antialiasing oscillators for subtractive synthesis [3]. BLAMP corner rounding treats discontinuities in the first derivative (triangular edges, clipping, rectification) without heavy oversampling [2]. We score and illustrate cycle-local approximations of those three operators beside DualCosine on the same prolonged- R protocol (Figure 5, Section 5.7). They are no longer citation-only anchors. DDSP reframes DSP modules as composable blocks inside learning pipelines [6]. We use that framing for small bake/FIR/MLP seam cells. We do not train end-to-end neural vocoders for every meta trial. These works are **used** as domain anchors for the operator vocabulary, as scored classical controls, and for why prolonged cyclic playback is the outer judge.

2.2 Label-free restoration

Noise2Noise shows that restoration mappings can be learned from corrupted observations alone by exploiting structure in the corruption process [4]. Speech Noise2Noise denoisers apply the same idea to waveforms without clean paired audio [5]. Mixture invariant training ranks unsupervised separation systems without stem labels [29]. That philosophy motivates ranking denoise candidates without paired studio-clean wavetable cycles. Earlier drafts of this work cited Noise2Noise only as ranking philosophy and omitted a same-domain learned control. We now train a true corrupt $\rightarrow$ corrupt N2N-style seam restorer on the same $L=256$ `make_batch` geometry (primary baseline), plus a sibling-supervised ceiling and lightweight LSTM / 1D-CNN seq models (Section 5). Prior speech N2N restorers operate on broadband noisy speech. We operate on periodic seam cliffs with procedural siblings. We **use** this family as conceptual justification for unsupervised residual ranking and as an explicit learned contrast to DenoiseOpt meta-search. Full Demucs / DiffWave stacks are screened: domain mismatch (full-band speech/music denoisers vs short periodic seams), compute, and OA/eval protocol mismatch. We do not retrain image Noise2Noise CNNs.

2.3 Waveform denoising and separation cells

Deep learning for audio surveys the move from hand-designed features to learned time/frequency models [7]. Wave-U-Net performs end-to-end source separation with multi-scale 1-D U-Nets [8]. Improved Wave-U-Net speech enhancement and singing-voice U-Nets popularized encoder-decoder skips on audio [9, 10]. The original U-Net, ResNet residuals, and Transformer attention supply the block primitives we shrink to bake width [13–15]. Conv-TasNet and dual-path RNN demonstrated strong time-domain separation with temporal convolutions and long-context paths [11, 12]. SepFormer and audio spectrogram transformers push attention-heavy audio stacks further [30, 31]. These lines are **used** as design priors

for tiny searchable cells (shallow U-Net / dilated / dual-path / attention blocks under a strict per-trial bake budget). They are not frozen full-scale separators inside every meta trial.

2.4 NAS and RL controllers

Elsken et al. taxonomize NAS by search space, strategy, and evaluation cost [16]. Their performance-estimation axis includes lower-fidelity proxies such as shorter training, data subsets, and lower-resolution images [16]. Zoph and Le couple RL controllers to architecture search [17]. ENAS adds parameter sharing for efficient discrete search [18]. Progressive and platform-aware NAS add staged evaluation and latency-aware rewards [27, 28]. We **use** this family as the ancestor of RL proposing architecture edits (add/remove/mutate ops, toggle residual-primary, mutate MLP/FIR), scored with prolonged residual. PPO supplies the clipped-surrogate policy update when the hybrid search loop logs PPO mutation branches [19].

2.5 Evolutionary NAS and population schedules

Large-scale and regularized / aging evolution showed that evolutionary NAS can discover competitive image classifiers when population turnover is carefully regularized [20, 21]. Population Based Training jointly adapts weights and hyperparameters in an asynchronous population [22]. Practical Bayesian optimization [26] and CMA-ES tutorials [24] inform local Bayesian and continuous evolutionary prior families. These are **used** for GA tournament/crossover branches, PBT-style exploit-mutate, and literature-combo priors, implemented at synthesizer-trial scale.

2.6 Evolutionary reinforcement learning hybrids

Khadka and Tumer’s evolution-guided policy gradient (ERL) interleaves an evolutionary population with an RL actor to mitigate sparse reward and brittle hyperparameters in deep RL [23]. We **use** ERL as the spirit of interleaving GA population steps with PPO updates at tiny population sizes. We do not claim a full ERL robotics stack.

2.7 Mixture-of-experts soft gating

Sparsely gated MoE layers increase capacity via conditional computation [25]. We **use** MoE as motivation for soft gates over heterogeneous seam blocks (e.g., U-Net vs dilated vs attention cells) under residual scoring, at bake-scale widths.

2.8 Used versus screened

Used (methodological or comparative anchors).

- Discontinuity / VA lineage: [1–3].
- Unsupervised restoration philosophy: [4, 5, 29].
- Modular / differentiable audio DSP context: [6].
- Audio DL and compact waveform cells: [7–15, 30, 31].
- NAS / RL controllers / PPO: [16–19, 27, 28].
- Evolutionary NAS and population HPO: [20–22, 24, 26].

- ERL hybrid outer loop: [23].
- MoE soft gating prior: [25].

Screened (contrast only).

- **MAML** [32]: bi-level / fast-adaptation framing. We do not differentiate through a deep task network. Inner fits are coordinate-style updates on bake parameters. Outer rank is residual R .
- **DARTS** [33]: continuous relaxation of architecture search. We keep discrete ops + small cells under residual scoring.
- **Demucs / Hybrid Demucs / realtime waveform enhancement** [34–36]: strong separators and enhancers, but full multi-scale / hybrid STFT stacks are too heavy per meta trial under bake-cell search budgets.
- **DiffWave** [37]: diffusion vocoder. Screened as a generative sampling stack outside the seam bake operator set.
- **SEGAN-style adversarial enhancement** [38]: optional tiny adversarial cues only. Full GAN training loops screened for per-trial cost.

Positioning. Relative to this map, the present work contributes (i) a prolonged residual objective for periodic seam denoising, (ii) an explicitly hybrid RL+GA(+PBT) meta-outer loop with honest branch naming, and (iii) a used/screened literature protocol with downloaded source PDFs. Claims stay inside cycle-local seam restoration under a declared compute budget. The primary claim is protocol-level: residual-scored meta-search for cycle-local audio denoise operators.

3 Methods

3.1 Notation

Throughout, $x \in \mathbb{R}^L$ is the cracked single-period wavetable (engine input with open-wrap cliff). Let $\mathcal{H}$ be the space of bake-cell configurations (discrete operator graph plus continuous bake parameters). The bake operator is

$$\Theta : \mathbb{R}^L \times \mathcal{H} \rightarrow \mathbb{R}^L, \quad y = \Theta(x; h), \quad h \in \mathcal{H}. \quad (1)$$

When continuous parameters alone are emphasized we write θ for the continuous slice of h and $y = \Theta(x; \theta)$. $r^* \in \mathbb{R}^L$ is the procedural ideal sibling defined below. Tiling length N (typically $N=16$) forms $y_{\text{tiled}} = \text{Tile}(y, N)$ and $r_{\text{tiled}}^* = \text{Tile}(r^*, N)$. Seam width W (code: SEAM_W, typically 8) is the neighborhood edited by classical fade and polish ops. $R \in [0, 1]$ is the prolonged residual score (1 = best). These symbols follow a different convention from Noise2Noise clean/noisy pairs.

3.2 Problem and ideal sibling

Problem. Given a cracked period x , choose a bake configuration h so that the repaired cycle $y = \Theta(x; h)$ matches a no-cliff ideal under prolonged tiling.

Table 1: Core symbols used in Methods.

Symbol	Meaning
L	Cycle length (samples per period)
x	Cracked engine wavetable period (cliff on)
$y = \Theta(x; h)$	Baked cycle after seam ops / residual cell
Θ	Bake operator $\mathbb{R}^L \times \mathcal{H} \rightarrow \mathbb{R}^L$
$h \in \mathcal{H}$	Bake-cell configuration (ops + continuous θ)
r^*	Ideal sibling (same seed; cliff withheld)
G	Procedural period generator (cliff on/off)
N	Prolong tiling count
W	Seam neighborhood width
R	Prolonged residual score in $[0, 1]$ (1 = best)

Ideal sibling (Q1). Let $G(\text{seed}, \text{cliff})$ denote the procedural generator used by the CUDA FitCell batch maker (`make_batch` in `overnight_gpu_rl_arch.py`): one draw of frequency, phase, and mid-cycle content, with an optional open-wrap cliff of width W at both ends. For any seed,

$$r^* = G(\text{seed}, \text{cliff}=\text{off}), \quad x = G(\text{seed}, \text{cliff}=\text{on}). \quad (2)$$

Engine and ideal therefore share the mid-cycle content of one draw and differ by the seam cliff (plus any seam-boosted noise applied only on the engine path). The ideal is *not* a latent studio ground truth and is *not* a method output. For an arbitrary cracked period from this family, the sibling is obtained by replaying the same seed with the cliff withheld—exactly the pair returned by `make_batch`. This sibling relationship makes unsupervised ranking possible without paired studio recordings [4, 5].

3.3 Outer objective and residual score R

Single outer objective. Every candidate bake—including classical controls and searched cells—is ranked by maximizing absolute prolonged residual against the same ideal sibling:

$$\max_{h \in \mathcal{H}} R(y, r^*) \quad \text{subject to} \quad y = \Theta(x; h). \quad (3)$$

Equivalently, FitCell minimizes loss $1 - R$. Selection, population ranking, and champion updates use absolute R only.

Every scored row—no-bake, DualCosine, FIR, poly, VA residuals, neural favorite—applies some Θ to the cracked engine x , then compares tiled y to the *same* r^* under (4). The ideal sibling is the scoring target; it is not a method and not the no-bake baseline.

With tiling length N ,

$$R = \text{clamp} \left(1 - \frac{\text{rms}(y_{\text{tiled}} - r_{\text{tiled}}^*)}{\max(\text{rms}(r_{\text{tiled}}^*), \varepsilon)}, 0, 1 \right), \quad (4)$$

where $\text{rms}(z) = \sqrt{\frac{1}{|z|} \sum_i z_i^2}$ and $\varepsilon > 0$ avoids division by zero. By construction $R \in [0, 1]$, $R=1$ when $y_{\text{tiled}} = r_{\text{tiled}}^*$, and (when pre-clamp scores lie in $(0, 1)$) R is strictly decreasing in residual RMS for fixed ideal RMS. A method “beats” another iff its R is higher on the same (x, r^*) draw. Optional soft shape gates

(when enabled) multiply R by factors that down-weight mid-cycle damage relative to seam-local error. We do not claim convergence guarantees for the hybrid outer loop, nor a general wrap-closure theorem for arbitrary bake cells. Outer-loop cost scales as population size $\times$ proposals per iteration $\times$ fit steps $\times$ batch forward cost (order 10^3 – 10^4 arch evaluations at the reported 5k gate).

Remark 1 (Noise2Noise motivation). Ranking without studio-clean pairs is possible when the ideal is a procedural sibling of the engine draw [4, 5].

3.4 Bake operator Θ and classical baselines

A **bake cell** $\Theta(\cdot; h)$ composes classical seam ops (DualCosine, polish, soft seam, FIR blends) with optional tiny residual cells (MLP/FIR/U-Net-lite/attention-lite/LSTM-lite) and optional MoE soft gates over experts (Appendix A, Algorithms 2–4). Bake parameters live in a bounded box; architecture strings are discrete graphs over those ops. Separate classical controls apply cycle-local BLIT/BLEP, PolyBLEP, and BLAMP residual corrections at the wrap on the cracked engine tile (same interface as DualCosine / FIR; not a full oscillator rewrite). Plain-language roles: BLIT/BLEP and PolyBLEP correct a *value* step at wrap; BLAMP corrects a *slope*/corner jump; DualCosine fades endpoints without a BLEP residual. They are scored in Section 5.7.

Four roles (do not conflate).

- **Ideal sibling** r^* : cliff withheld; scoring target only (panel (a) in Figure 1).
- **No-bake** (passthrough): $\Theta_{\text{no-bake}}(x) = x$; unrepaired cracked engine scored vs r^* (panel (b)). Not r^* , not a residual-net skip, not a learned identity map.
- **DualCosine**: classical raised-cosine end fade on x (panel (c)); one classical board row and, separately, a PPO advantage baseline (Remark 2).
- **Ours**: searched bake $y = \Theta(x; h^*)$ from the hybrid outer loop (panel (d)).

On mild cliffs residual mass $\ll$ ideal RMS under (4), so no-bake R can sit near 0.97 while the wrap click remains audible—a “do nothing” reference, not a perceptual wrap score. Released JSON may still use the legacy key *identity* for no-bake; manuscript prose uses *no-bake*.

Classical (non-AI) comparison board. Learned / searched methods are always ranked against the full classical board on absolute R (same r^*): no-bake, DualCosine / cosine fade, *seam_fir3*, classic quadratic, hann/crossfade/soft fades, detrend/ensemble, poly seam, and cycle-local BLIT/BLEP / PolyBLEP / BLAMP where scored. When tables show ΔR vs DualCosine, that is a reporting gap vs one classical bake, not the definition of reward.

Remark 2 (DualCosine centering is PPO advantage only (Q2)). The outer objective remains (3): maximize absolute

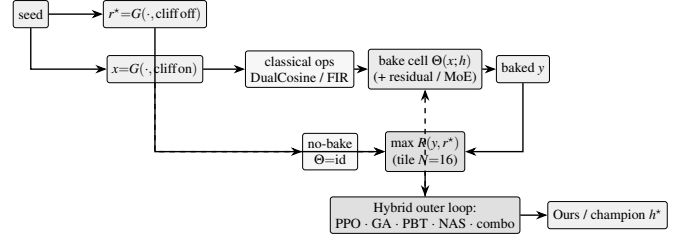


Figure 2: DenoiseOpt overview (before component deep-dives). Same-seed generator G yields ideal sibling r^* (cliff off) and cracked engine x (cliff on). Bake cell $\Theta(x; h)$ may include DualCosine/FIR and optional residual/MoE experts; no-bake is the identity control $x \mapsto x$. Outer objective is maximize absolute prolonged $R(y, r^*)$; DualCosine centering (not shown) is PPO advantage shaping only. Dashed feedback: architecture proposals update the cell. Pseudocode in Appendix A.

$R(y, r^*)$. In the PPO branch only, we optionally form a centered advantage

$$\rho = R - R_{\text{DualCosine}}, \quad (5)$$

so early-search advantages are roughly zero-mean. This does *not* change selection, FitCell loss, champion ranking, or the matched 5k meta-approach objective—all of which still maximize absolute R . DualCosine centering is soft relative shaping for the actor–critic, not a hard absolute ranking target.

3.5 Pipeline overview

Figure 2 summarizes the residual-scored hybrid loop before component detail. Each outer iteration proposes a bake configuration h , runs FitCell (Appendix A, Algorithm 5) to fit continuous parameters under loss $1 - R$, evaluates prolonged R vs r^* , and updates the population / PPO policy. Pseudocode for GA, PPO, PBT, and the full hybrid rotation is in Algorithms 6–9.

3.6 Search space and FitCell

Each genome encodes: ordered op list (fade, polish, FIR, pin, residual write), depth, residual cell type (none, MLP, FIR, U-Net-lite, attn-lite, LSTM-lite), optional MoE expert count $K \in \{2, 3, 4\}$, and continuous θ in the CUDA runner box bounds. The searchable bake vocabulary includes a width-capped 1–2 layer LSTM block over the seam window or full cycle (distinct from the fixed supervised seq-LSTM SOTA baseline). The live hybrid tag is PPO+GA+PBT+NAS+depth+MoE with LSTM enabled in the block graph.

FitCell draws sibling batches via G , applies $\Theta(\cdot; h)$, and steps Adam on $1 - R$ with early stopping on plateau (Algorithm 5).

3.7 Hybrid outer loop

Each iteration rotates among branches: PPO, GA, PBT, NAS, and combo. Selection and FitCell are always by absolute residual R (maximize toward the ideal sibling; loss $1 - R$).

Near-ceiling R and why reward / search HPs matter. On the sine+cliff generator, no-bake already sits near

$R \approx 0.97$, and strong classical / searched bakes live in a narrow band toward 1 (often 0.99–0.991). Absolute gaps of order 10^{-3} – 10^{-2} are scientifically real but tiny as raw PPO/GA signals: without shaping, advantages collapse and hyperparameters (clip, entropy, mutation rate, population size, fit steps, learning rate) dominate whether the outer loop can climb the last percent. We therefore treat **reward shaping and search hyperparameters as first-class tunables** alongside bake architecture. In the present hybrid, PPO uses a searchable shaping mode among `{abs_r, vs_dualcosine, vs_nobake, neglog_gap}` (PBT-mutated; default $\rho = R - R_{\text{DualCosine}}$ per Remark 2). PBT mutates fit / PPO hyperparameters and reward mode in-population; plateau adaptation retunes branch mix and depth/MoE bias when champions stall. The matched 5k meta-approach comparison (Section 3.8) is the sole GPU experimentation vehicle for outer-loop family, HP, and reward-shaping sweeps under a shared FitCell budget.

3.8 Meta-learning approach comparison

Beyond the hybrid outer loop, we compare five additional outer-loop families under a matched 5,000-evaluation budget, shared search seed 1902771841, and the same FitCell / prolonged- R protocol (LSTM and xLSTM included in the searchable bake vocabulary). Table and figure legends use the short display names below (artifact folders keep stable code keys; see NOMENCLATURE.md):

- **Random NAS** (`random`): independent uniform draws over bake graphs and fit hyperparameters (control).
- **Cont. CMA-ES** (`cmaes`): diagonal covariance-matrix adaptation over a continuous encoding of depth, width, cell kind, block logits, and fit hyperparameters [24].
- **Arch REINFORCE** (`reinforce`): vanilla policy gradient over discrete architecture-edit actions (no clipped surrogate; contrast to PPO [17, 19]).
- **Aging evolution** (`aging_evo`): regularized evolution with oldest-member replacement after tournament parent selection [20, 21], distinct from the hybrid’s non-aging tournament GA branch.
- **TPE Bayes NAS** (`tpe`): lightweight tree-structured Parzen estimator over discrete bake categoricals in the spirit of practical Bayesian optimization [26].

We also run **Ours (hybrid GA-PPO)** (`hybrid_lstm`): the multi-branch hybrid loop for 5,000 evaluations with LSTM/xLSTM in the block vocabulary and online reward-mode / fit-PPO hyperparameter co-tuning, as the proposed matched-budget arm. Results appear in Table 12 and Figures 6–7; healed-sample overlays in Figure 8.

Remark 3 (Trial complexity). Per outer iteration the dominant cost is $O(T_{\text{fit}} \cdot C_{\text{fwd}})$ for fit steps times one forward bake. Bake-width caps (tiny residual cells) keep C_{fwd} far below full-band Demucs/DiffWave forwards screened in Related Work.

Table 2: Hybrid search hyperparameters (RTX 3090 DenoiseOpt CUDA search runner / `denoise_meta_evo.py` defaults). Co-tuned with reward shaping under near-ceiling R (Section 3.7).

Parameter	Value
Population size n	12
GA tournament k	3
GA crossover fraction p_c	0.5 (else clone elite parent)
GA mutation	≥ 1 mutate after inherit (+50% second mutate)
PPO clip ϵ	0.2 (default; searchable <code>{0.1, 0.2, 0.25, 0.3}</code>)
Adam learning rate (fit)	3×10^{-3} (default; PBT-searchable)
Adam learning rate (policy)	3×10^{-4}
Entropy coefficient (PPO)	0.02 (default; searchable ≈ 0.005 –0.06)
PBT exploit / mutate	elite fraction 0.25; multi-step arch+hp mutate
MoE routing	<code>sequential</code> or <code>moe_parallel</code> when enabled
Plateau boredom threshold	1000 iters without champ improve
Cycle length L / tiles N / seam W	256 / 16 / 8
Fit steps / batch (default)	24 / 48
Algo tag	PPO+GA+PBT+NAS+depth+MoE

3.9 Hyperparameters

Table 2 lists the CUDA search defaults used for the 5k-gate freeze reported in Results. Under the near-ceiling residual regime (Section 3.7), these outer-loop and fit hyperparameters are co-tuned with architecture; PBT and plateau adaptation continue to mutate several of them online. Companion listings: Algorithms 1–9 (Appendix A) and `docs/PSEUDOCODE.md`.

4 Experiments

Protocol freeze. Evaluation follows `paper/v7/EVAL_PROTOCOL.md` (v1, extended for peer-review strata). Primary metric: prolonged R . Secondary: SNR, SDR, wrap-jump (diagnostic), and required seam-local edge RMSE on tiled audio. Holdout seed 20260719. Search seed 1902771841. We report the 5k-gate freeze plus frozen holdout. We make no unfinished-budget claims. PESQ/STOI stay out of scope on non-speech cycles. LibriSpeech and MUSDB are not used. Venue positioning: DSP / synthesis artifact repair (DAFx / AES / arXiv cs.SD), not general speech-enhancement leaderboards.

4.1 Dataset

Creation. The paper-facing evaluation corpus is a frozen draw from the same synthetic generator used by the CUDA search runner (`make_batch` in `overnight_gpu_rl_arch.py`). Each cycle has length $L=256$. A two-harmonic sine is drawn with frequency

$\sim \mathcal{U}(1,4)$ and phase $\sim \mathcal{U}(0,2\pi)$. An open-wrap seam cliff of amplitude $\pm \mathcal{U}(0.08,0.43)$ is applied over $\text{SEAM_W}=8$ samples at both ends. Seam-boosted Gaussian noise (scale 0.02 at the ends, $0.15\times$ in the mid-cycle) is added to form the engine input. The ideal withholds the cliff and is used only as the scoring target r^* (never as a method output). No-bake leaves the cracked engine unchanged and scores that unrepaired x against the same r^* . Prolonged residual R tiles each cycle $N=16$ times before the RMS ratio. The holdout seed is 20260719 (distinct from the hybrid search seed 1902771841). The search campaign draws fresh i.i.d. batches from this generator and does not train on the frozen holdout tensors. There is no supervised clean/noisy training split of studio audio.

Multi-family diversity (≥ 20 waveforms). Beyond the single sine+cliff holdout, Phase 3a scores methods on **20** generative draws: ten family variants (sine_cliff, harmonic_fft, am_fm, nonlinear, combo, triple_mix, extreme_overlay, open_wrap_bias, soft_noise, wide_seam) $\times$ two seeds (20260719 and 20260720), batch 64 each. These Python generative variants stress harmonic / AM-FM / nonlinear / overlay / open-wrap behavior of make_batch and remain the primary SOTA matrix. Separately, export_sound_bench_tiles exports **20** Rust sound_bench engine/ideal pairs (10 families $\times$ 2 seeds, $L=256$, byte-aligned with generate_sound / generate_sound_ideal) scored under the same residual geometry as a secondary transfer block (Table 13). Every method sees the same tensors per waveform. **Primary classical comparison** is absolute prolonged R against the non-AI board (no-bake, DualCosine, seam_fir3, poly, fades, VA residuals; Section 3.4). The search objective is maximize R toward the ideal sibling (best $R=1$). DualCosine is one classical comparator and a PPO centering constant; it is not the reward target.

Dataset metrics. A metrics draw of 4,096 cycles under the holdout seed yields mean ideal RMS 0.721 ± 0.016 , mean engine residual RMS 0.021 ± 0.013 , and mean absolute wrap jump $|x_0 - x_{L-1}|$ of 0.913 ± 0.634 . No-bake (passthrough) on that draw scores mean $R \approx 0.971$. Method tables use a matched score batch of 64 cycles from the same seed (tensors under brand/artifacts/canonical_eval_dataset/), five consecutive seeds for mean $\pm$ std on the classical table, and the 20-waveform matrix for SOTA secondary metrics. Figure 3 summarizes the 4,096-cycle metrics draw: ideal RMS clusters near 0.72, wrap jumps span nearly three orders of magnitude (median ≈ 0.83), and no-bake R is high on easy tiles but falls below 0.91 on the hardest decile of wrap jumps. Figure 1 is the documented sine-wrap edge case: a high-|wrap| holdout tile where cliff modulation is visible, scored with DualCosine and the top neural favorite under the same residual R (tile $R \approx 0.9917$, favorite batch mean $R \approx 0.991$).

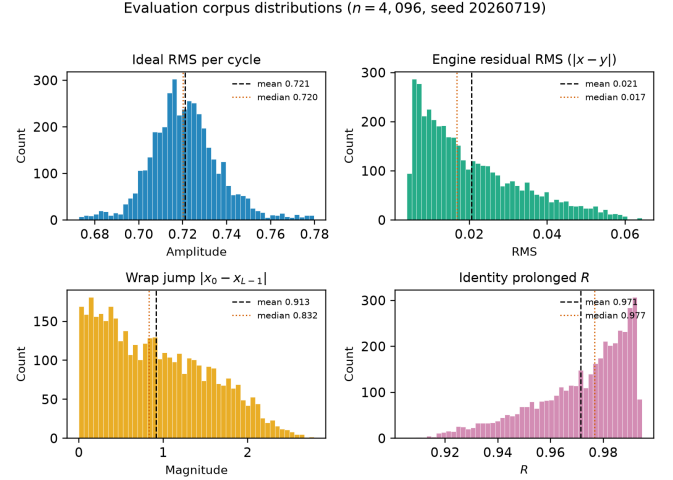


Figure 3: Evaluation corpus statistics on the frozen holdout seed (20260719, $n=4,096$ cycles). Distinguish: ideal sibling r^* (cliff withheld); no-bake = unrepaired cracked engine (not r^*); DualCosine = one classical bake. Top left: ideal RMS per cycle. Top right: engine residual RMS before any bake. Bottom left: wrap jump magnitude $|x_0 - x_{L-1}|$ (cliff hardness proxy). Bottom right: no-bake prolonged R (passthrough, $x \mapsto x$). Dashed and dotted vertical lines mark mean and median. Hard-cliff reporting (Section 5) uses the top 10% wrap-jump stratum where no-bake R drops.

Hybrid search campaign (5k clean gate). A CUDA search tagged PPO+GA+PBT+NAS+depth+MoE runs with a large declared iteration budget (order 10^5). At the reported freeze (run gpu-rl-arch-20260719T083019Z, $\approx 6,922$ clean iters / ≈ 7.9 h on RTX 3090) the champion residual is $R \approx 0.99093$ against DualCosine baseline ≈ 0.81658 ($\Delta R \approx +0.174$). Branch-best residuals at the freeze: GA ≈ 0.99016 , PBT ≈ 0.99012 , PPO ≈ 0.98924 , combo ≈ 0.98854 , NAS ≈ 0.98677 . Isolated short-budget re-runs (GA-only, PPO-only, GA+PPO, full, 150 iters, same search seed, plateau adapt off) are reported beside those freezes in Table 10. Figures in Section 5 regenerate from history.jsonl near this gate.

Inference, classical, and learned baselines. High- R fitted cells ($R \geq 0.98$) are timed on CUDA (batch 64, prolonged residual). Ten non-learnable bake denoisers share the frozen canonical holdout against the neural favorite (Table 4). Cycle-local BLIT/BLEP, PolyBLEP, and BLAMP seam repairs are scored on the same holdout / cliff / multi-family protocol (Table 9; va_seam_blep.json) and illustrated on the intro high-wrap tile (Figure 5). Fixed-arch MLP-on- R and CNN/U-Net-lite-on- R baselines (no outer NAS) are trained on $1-R$ with the same generator and scored in the SOTA matrix. Phase B adds a primary corrupt $\rightarrow$ corrupt Noise2Noise seam CNN (train seed 424242, disjoint from holdout), a secondary sibling-supervised ceiling on the same architecture, and lightweight LSTM / depthwise 1D-CNN seq restorers (train seed 424243). No search-champion weights initialize

those baselines. Family-conditional hardness uses the separate Rust procedural 100,000-cycle bench reported in Results.

Cliff strata and wavetable-native realism. A 4,096-tile holdout draw (seed 20260719) is stratified by engine wrap-jump percentiles (top 25% / top 10%; cutoffs frozen in `cliff_strata.json`). Hard-cliff is the operative regime for claims about DualCosine collapse vs favorite retention. Edge RMSE is the locked required seam-local secondary; click energy is optional. Wavetable-native transfer (Phase F1) scores true ReelSynth-exported factory periods (primary) and external AKWF CC0 WAVs (secondary) under the same close-seam→open-wrap protocol. Procedural stand-ins are tertiary smoke only. Speech/music LibriSpeech/MUSDB corpora remain off.

Audible demos and vibrato spectrograms. Downloadable wavetable clips (no-bake / DualCosine / Ours) live under `reelsynth/brand/artifacts/meta_approach_comparison/` (44.1 kHz mono, A4, 3s; rebuild: `scripts/export_meta_hear_samples.py`). A slow-vibrato spectrogram protocol (`scripts/bench_vibrato_spectrogram.py`) scores the same bake cells under modulated-pitch playback; figures appear in Section 5.12. These assets support informal audition and spectrogram inspection only—formal listening panels are out of scope (Section 9).

5 Results

Family-conditional hardness (completed bench). RQ: which procedural families dominate remaining residual mass? On the fitted DenoiseOpt 100,000-cycle bench, denoise score D is lowest for nonlinear ($D \approx 0.39$) and combo ($D \approx 0.40$), then `triple_mix` / `extreme_overlay` ($D \approx 0.51$ – 0.52), while `harmonic_fft` and `am_fm` sit near $D \approx 0.69$ – 0.70 . Lit-combo champion family stress (prolonged residual) similarly ranks `open_wrap_bias` ($R \approx 0.83$) and `extreme_overlay` / `combo` ($R \approx 0.88$ – 0.89) below `harmonic_fft` ($R \approx 0.95$). Hard sounds recur.

Hybrid search (5k-gate freeze). RQ: what champion R does the hybrid search campaign reach at the clean 5k-iteration gate? At the reported 5k-gate search freeze ($\approx 6,922$ clean iterations, ≈ 7.9 h on RTX 3090) the champion residual is $R \approx 0.99093$ with DualCosine ≈ 0.81658 on the CUDA search runner ($\Delta R \approx +0.174$). Monitoring plots below are regenerated from that freeze. Declared larger budgets continue. This paper reports the gate freeze only.

Inference time at matched score. RQ: among near-max- R fitted cells, which is fastest at matched score? Across diverse high- R fitted cells ($R \geq 0.98$), we timed CUDA forward passes (batch 64, prolonged residual). Within a tight band of the best residual ($R \geq R_{\max} - 5 \cdot 10^{-4}$), the **favorite** is the lowest-latency model: `champion_iter_000235` with $R \approx 0.9910$, ≈ 3.15 ms/batch, ≈ 37 k parameters (versus a higher-capacity max- R cell at $R \approx 0.9914$ but ≈ 7.18 ms/batch

Table 3: Top-5 distinct fitted architectures on the CUDA inference bench (batch 64, prolonged R , seed geometry matching the search campaign). Favorite is latency-best inside the near-max- R band ($R \geq R_{\max} - 5 \cdot 10^{-4}$).

Tag	R_{saved}	R_{live}	ms/batch	params
<code>iter_000157</code>	0.9914	0.9913	7.18	104 484
<code>iter_000205</code>	0.9911	0.9907	7.29	70 562
<code>iter_000235*</code>	0.9910	0.9915	3.15	37 489
<code>iter_000397</code>	0.9910	0.9907	6.80	97 262
<code>iter_000229</code>	0.9909	0.9902	6.67	100 500

*Favorite (fastest in $R \geq R_{\max} - 5 \cdot 10^{-4}$ band).

/ ≈ 104 k params). Selection rule: maximize R , then minimize latency inside the band.

5.1 Top-5 architectures

RQ: which fixed bake graphs achieve near-max R at lowest latency on the inference bench? Table 3 ranks five distinct fitted bake graphs from the CUDA inference bench (`inference_bench.json`, batch 64), preferring residual R first and latency on ties, and skipping near-duplicate copies of the same topology signature.

Classical vs favorite (canonical holdout). Table 4 scores the classical (non-AI) bake set—including no-bake, DualCosine, and `seam_fir3`—plus the neural favorite on the frozen corpus of Section 4.1 (CUDA, batch 64, seed 20260719). Every row uses the same ideal sibling r^* ; higher R means closer to that sibling. DualCosine reaches $R \approx 0.825$ (≈ 1.35 ms/batch), near the search-monitor DualCosine ≈ 0.817 on live draws from the same generator. The best *active* classical repair is seam-local 3-tap FIR (`seam_fir3`) at $R \approx 0.961$ / ≈ 0.67 ms/batch (0.962 ± 0.002 over five seeds). No-bake (passthrough; unrepaired engine vs r^*) scores $R \approx 0.972$ and is a do-nothing reference, not the ideal. The neural favorite reaches $R \approx 0.991$ (0.991 ± 0.0002 over five seeds) at ≈ 2.56 ms/batch (≈ 37 k params): $\Delta R \approx +0.019$ vs no-bake, $\Delta R \approx +0.030$ vs `seam_fir3`, and $\Delta R \approx +0.166$ vs DualCosine.

5.2 SOTA matrix (multi-family + secondary metrics)

RQ: does the meta favorite beat the classical non-AI board (no-bake, DualCosine, FIR, poly, fades) and fixed MLP/CNN baselines across twenty generative families? Table 5 reports mean $\pm$ std over the 20-waveform multi-family set (Section 4.1), with canonical-holdout SNR/SDR/wrap-jump for the same method set from `sota_matrix.json`. PESQ/STOI/MUSHRA are omitted for non-speech cycles (Limitations). Primary reading is absolute R rank vs classical rows (no-bake 0.965, poly 0.966, `seam_fir3` 0.955, DualCosine 0.814). Fixed MLP-on- R and CNN/U-Net-lite-on- R (no outer NAS) sit between DualCosine and the meta favorite. Paired Wilcoxon signed-rank (normal approx.) of favorite vs DualCosine on the 20 waveforms: $W=0$, $p \approx 8.9 \times 10^{-5}$, mean $\Delta R = +0.162$ (bootstrap 95% CI [0.155, 0.170]); vs no-bake

Table 4: Method scores on the canonical sine+cliff holdout (batch 64, seed 20260719, $L=256$, $N=16$). Mean $\pm$ std over five consecutive seeds. Every method is scored vs the same ideal sibling r^* . No-bake = unrepaired engine (not r^*). Rank by absolute R against the classical board; DualCosine is one classical row, not the sole comparator. Primary metric is prolonged R (1 = best).

Method	R	R (5-seed)	ms/batch
neural favorite	0.9911	0.9912 ± 0.0002	2.56
no-bake (passthrough)	0.9718	0.9726 ± 0.0021	0.31
seam_fir3	0.9610	0.9616 ± 0.0019	0.67
classic quadratic	0.8449	0.8453 ± 0.0117	1.59
DualCosine	0.8249	0.8258 ± 0.0137	1.35
cosine fade	0.8249	0.8258 ± 0.0137	1.59
crossfade	0.8116	0.8130 ± 0.0145	1.50
hann blend	0.8048	0.8060 ± 0.0156	0.54
soft wide fade	0.7432	0.7378 ± 0.0171	2.64
detrend	0.4079	0.4091 ± 0.0386	0.36
ensemble detrend+DC+FIR	0.4024	0.4027 ± 0.0387	1.56

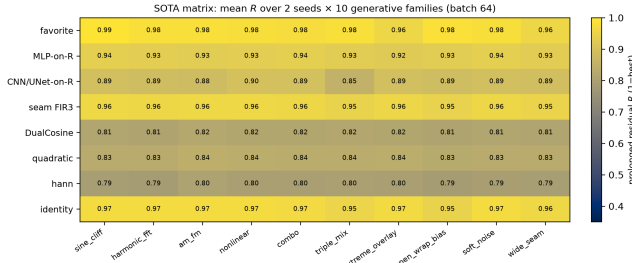


Figure 4: Colorblind-safe (cividis) heatmap of mean prolonged R for selected methods $\times$ generative families (2 seeds each). Higher is better.

the multifamily mean gap is smaller ($\approx +0.012$) but still favors the searched bake on this generator.

5.3 Hard-cliff strata

RQ: does high no-bake R hide DualCosine failure on large wrap jumps? Table 6 stratifies a 4,096-tile holdout draw (seed 20260719) by engine wrap-jump percentiles ($p75 \approx 1.387$, $p90 \approx 1.832$; frozen in `cliff_strata.json`). No-bake R barely moves (all $0.9715 \rightarrow$ top-10% 0.9667) because residual mass remains small relative to ideal RMS under (4). DualCosine collapses on hard cliffs ($0.819 \rightarrow 0.657$), while the meta favorite holds ($R \approx 0.9915$). Favorite wrap-jump on the top-10% subset stays ≈ 1.83 (vs no-bake 2.09). The claim is tiled residual repair, not endpoint-zeroing. **Edge RMSE** is the locked required seam-local secondary (EVAL_PROTOCOL): on the top-10% subset it drops from no-bake 0.095 to favorite 0.021 (DualCosine 0.990). Click energy remains an optional diagnostic.

5.4 N2N and seq baselines

RQ: how does DenoiseOpt compare to true Noise2Noise-style and lightweight seq restorers on the same generator? Primary corrupt $\rightarrow$ corrupt N2N (20k Adam steps, 53k params, train seed 424242) reaches canonical $R \approx 0.9917$, matching the meta favorite within ≈ 0.001 . Secondary sibling-supervised N2N reaches $R \approx 0.9933$. A bidirectional LSTM (76k params, 15k steps, train seed 424243) is the supervised ceiling at $R \approx 0.9952$. A tiny depthwise 1D CNN (3.2k params) reaches $R \approx 0.9864$. Holdout tiles never enter training. Classical DualCosine / seam_fir3 remain in Table 6.

5.5 Wavetable-native transfer

RQ: does the wrap protocol transfer beyond synthetic sine cliffs onto *true* exports and third-party OA cycles? Table 8 scores ReelSynth-exported factory periods (primary realism, $n=25$; byte-true factory frames resampled to $L=256$) and Adventure Kid Waveforms (AKWF) CC0 single-cycle WAVs (secondary, $n=24$) after close-seam idealization and the same open-wrap cliff. Procedural stand-ins are demoted to tertiary smoke and no longer carry the primary realism claim. On exported factory periods the favorite ($R \approx 0.911$) trails no-bake / seam_fir3 / poly (domain shift) while still beating DualCosine (0.883). On AKWF OA cycles the favorite ($R \approx 0.970$) matches no-bake and beats DualCosine (0.936). LibriSpeech/MUSDB are not used.

5.6 Polynomial baseline and jump-aware control

RQ: does a cheap classical poly fitter close the S7 residual, and does endpoint pinning show the wrap-jump vs R tradeoff? A seam-local degree-3 polynomial fitter (no learning) reaches canonical $R \approx 0.972$ ($\approx$ no-bake) with wrap-jump essentially unchanged (`poly_baseline.json`). SSM seq models are deferred; the LSTM remains the supervised seq ceiling. An endpoint-pin control (`jump_control.json`) drives wrap-jump to 0 on all / hard subsets while dropping R (all 0.906; hard 0.822). Methods that zero wrap-jump are not the same objective as prolonged R ; we do not claim wrap-jump SOTA for the favorite.

5.7 Classical VA seam repairs (BLIT/BLEP, PolyBLEP, BLAMP)

RQ: do the cited VA antialias tools [1–3] beat DualCosine on prolonged R when applied as cycle-local seam residuals?

What each operator changes. Figure 5 shows the family on one high-wrap sine+cliff tile (same seed and tile as Figure 1). **BLIT/BLEP** (Stilson/Smith lineage) treats the wrap as a value step. It subtracts a band-limited step residual (here a raised-cosine BLEP-family lobe scaled by the measured wrap jump) so the tiled click spectrum is softened near the seam. **PolyBLEP** (Nam et al.) is a cheap polynomial surrogate of that step residual with fractional-delay support near phase wrap. It edits only a few samples on each side of the discontinuity. **BLAMP** (Esqueda et al.) corrects a slope or corner jump (discontinuity in the first derivative), which matters for triangles,

Table 5: SOTA comparison on 20 generative waveforms (10 families $\times$ 2 seeds, batch 64). R /SNR/SDR/wrap-jump are multi-family mean $\pm$ std. Rank by absolute R vs the classical non-AI board (no-bake, poly, FIR, DualCosine, ...). Column ΔR^* is vs DualCosine only (one classical gap for reporting; canonical freeze seed 20260719); ms* likewise. The search objective is maximize absolute R toward the ideal sibling, not DualCosine matching. Modern baselines (N2N/seq/poly) use disjoint train seeds (424242/424243); poly is classical (no train). Primary metric R (1 = best; closer to ideal sibling). No PESQ/STOI/MUSHRA.

Method	R (20-wave)	SNR (dB)	SDR (dB)	wrap-jump	ms*	params	ΔR^*
neural favorite	0.977 ± 0.010	35.0 ± 3.2	35.1 ± 3.2	0.90 ± 0.14	2.44	37 489	+0.166
N2N corrupt $\rightarrow$ corrupt	0.970 ± 0.021	33.9 ± 4.1	34.0 ± 4.1	0.90 ± 0.15	-	53 506	+0.167
seq CNN1D	0.969 ± 0.016	32.2 ± 3.5	32.4 ± 3.5	0.92 ± 0.15	-	3 242	+0.162
N2N sibling-sup.	0.969 ± 0.022	33.7 ± 4.3	33.8 ± 4.3	0.91 ± 0.15	-	53 506	+0.168
poly seam ($d=3$)	0.966 ± 0.008	31.2 ± 2.1	31.3 ± 2.1	0.91 ± 0.15	1.47	0	+0.148
no-bake	0.965 ± 0.008	30.9 ± 1.9	31.0 ± 1.9	0.91 ± 0.15	0.00	0	+0.147
seq LSTM	0.956 ± 0.037	32.2 ± 5.8	32.3 ± 5.8	0.86 ± 0.15	-	75 746	+0.170
seam_fir3	0.955 ± 0.005	28.1 ± 1.2	28.1 ± 1.2	0.46 ± 0.08	0.59	0	+0.136
MLP-on- R	0.932 ± 0.006	24.6 ± 0.9	24.6 ± 0.9	0.64 ± 0.07	2.76	8 604	+0.113
CNN/U-Net-on- R	0.886 ± 0.012	19.5 ± 0.9	19.5 ± 0.9	0.48 ± 0.06	5.01	11 752	+0.069
classic quadratic	0.835 ± 0.012	17.9 ± 1.1	17.7 ± 1.2	0.32 ± 0.05	1.78	0	+0.020
DualCosine	0.814 ± 0.013	16.9 ± 1.2	16.8 ± 1.2	0.32 ± 0.05	1.48	0	0
cosine fade	0.814 ± 0.013	16.9 ± 1.2	16.8 ± 1.2	0.32 ± 0.05	2.64	0	0
crossfade	0.802 ± 0.013	16.7 ± 1.2	16.4 ± 1.2	0.91 ± 0.15	1.70	0	-0.013
hann blend	0.793 ± 0.015	16.1 ± 1.2	15.8 ± 1.2	0.32 ± 0.05	0.57	0	-0.020
soft wide fade	0.729 ± 0.016	14.0 ± 1.1	13.6 ± 1.1	0.63 ± 0.08	3.69	0	-0.082
detrend	0.375 ± 0.040	5.5 ± 1.0	5.4 ± 1.0	0.00 ± 0.00	0.12	0	-0.417
ensemble DC+FIR	0.368 ± 0.040	5.1 ± 1.0	5.0 ± 1.0	0.12 ± 0.02	1.99	0	-0.422

Table 6: Cliff strata on 4,096 holdout tiles (seed 20260719). Strata by engine wrap-jump. Primary R (1 = best) and mean wrap-jump after bake. N2N primary = corrupt $\rightarrow$ corrupt; N2N secondary = sibling-supervised ceiling; seq = engine $\rightarrow$ ideal.

Method	all R	all jump	top25 R	top25 jump	top10 R	top10 jump
seq LSTM (ceiling)	0.9953	0.865	0.9955	1.637	0.9955	1.810
N2N sibling-sup.	0.9936	0.879	0.9936	1.668	0.9934	1.842
N2N corrupt $\rightarrow$ corrupt	0.9922	0.870	0.9924	1.647	0.9922	1.821
neural favorite	0.9911	0.869	0.9915	1.645	0.9915	1.825
seq CNN1D	0.9863	0.881	0.9869	1.684	0.9867	1.871
no-bake	0.9715	0.913	0.9713	1.797	0.9667	2.094
seam_fir3	0.9611	0.456	0.9461	0.893	0.9429	1.034
DualCosine	0.8193	0.335	0.6864	0.290	0.6574	0.335

n : all 4096, top25 1024, top10 410.

rectification, and hard clipping. On a pure value cliff the slope jump is small, so BLAMP is nearly a no-bake passthrough. **DualCosine** (our classical bake baseline) fades both ends of the stored period with raised-cosine windows. It does not insert a BLEP-style residual. It trades wrap energy for a wide end-fade distortion that prolonged R penalizes on hard cliffs.

Scores. Table 9 reports batch means from `va_seam_blep.json` (seed 20260719, batch 64). Implementation: `reelsynth/scripts/baselines/va_seam_blep.py` (PolyBLEP matches `osc::va::poly_blep`; BLIT/BLEP uses a raised-cosine step residual; BLAMP applies a poly-BLAMP lobe to the wrap slope jump). PolyBLEP reaches mean $R \approx 0.929$ ($\Delta R \approx +0.104$ vs DualCosine) but trails

no-bake / seam_fir3 / the meta favorite. BLIT/BLEP nearly zeros wrap-jump (0.004) while dropping mean R to 0.868 and collapsing on hard cliffs ($R \approx 0.712$). BLAMP is near-no-bake on this value-cliff geometry (mean $R \approx 0.972$). On the annotated severe tile in Figure 5, tile R is lower still (PolyBLEP ≈ 0.869 , BLIT/BLEP ≈ 0.703 , DualCosine ≈ 0.603 , BLAMP $\approx$ engine), which matches the hard-cliff strata. Honest reading: these classical tools are now scored, not only cited. They beat DualCosine on mean R in places. None matches the favorite on prolonged R .

5.8 Transfer conditions

RQ: when does hybrid search win vs the classical non-AI board (no-bake / FIR / DualCosine) and supervised nets? Per-cycle ΔR on export, AKWF, Rust `sound_bench`,

Classical VA seam techniques | seed=20260719, tile=46, |wrap|=2.283, L=256, periods=3

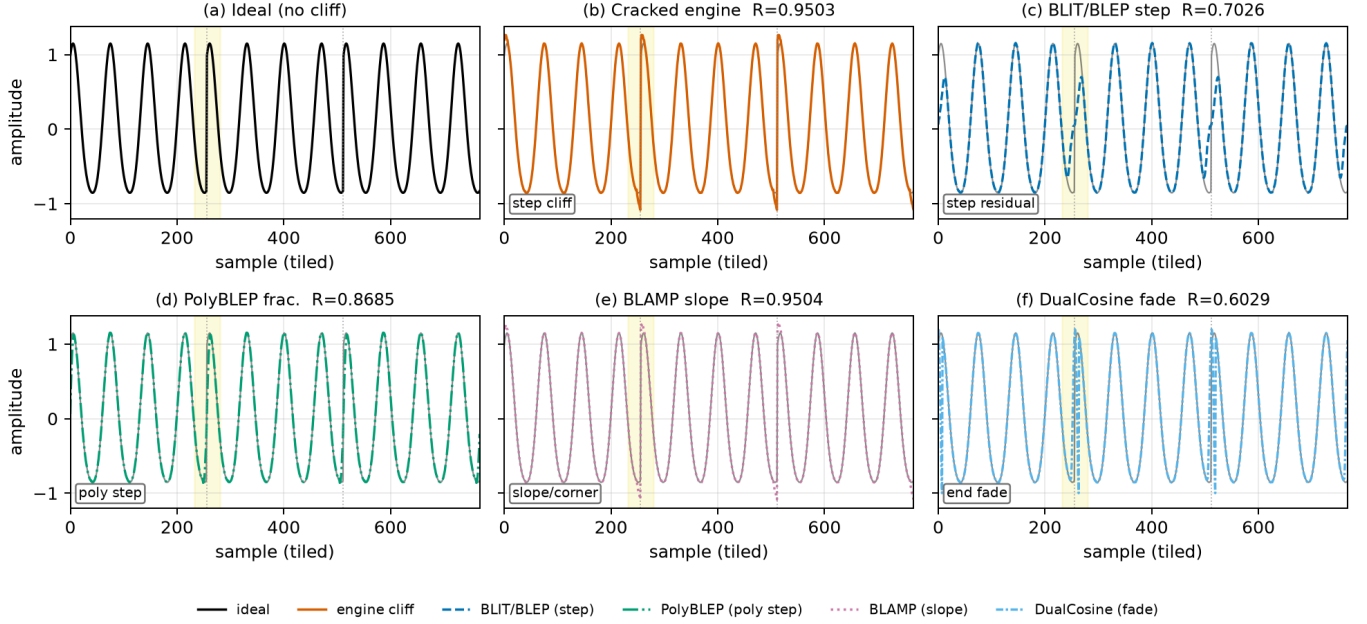


Figure 5: Classical VA seam techniques on one cracked sine+cliff tile (seed 20260719, tile 46, |wrap|=2.283, L=256, three tiled periods). Yellow band marks the first seam neighborhood. Panel tile R (1 = best): (a) ideal (unscored sibling). (b) cracked engine $R=0.9503$. (c) BLIT/BLEP step residual $R=0.7026$. (d) PolyBLEP fractional-delay residual $R=0.8685$. (e) BLAMP slope/corner residual $R=0.9504$ (near no-bake on value cliffs). (f) DualCosine end fade $R=0.6029$. Colorblind-safe hues with distinct linestyles for B&W print. Batch means appear in Table 9. Accessibility: six-panel comparison of ideal, cracked, BLIT/BLEP, PolyBLEP, BLAMP, and DualCosine.

Table 7: Required seam-local secondary on top-10% wrap-jump tiles ($n=410$): edge RMSE (lower better). Frozen in cliff_strata.json.

Method	top10 edge RMSE
seq LSTM	0.011
N2N sibling-sup.	0.018
neural favorite	0.021
N2N corrupt→corrupt	0.021
seq CNN1D	0.035
no-bake	0.095
seam_fir3	0.164
DualCosine	0.990

and sine+cliff holdout (transfer_failures.json): favorite win-rate vs DualCosine is high on sine cliffs (1.00) and AKWF (0.88), and moderate on export (0.64) / Rust (0.75); win-rate vs no-bake is high only on sine cliffs (0.92) and collapses on export (0.24) / Rust (0.20). Search value is therefore: hard cliffs, OA tracks where DualCosine collapses, and compact bake graphs without engine→ideal labels at search time. No-bake/FIR win on mild seams and OOD factory geometry (honest domain shift, not buried).

Table 8: Wavetable-native wrap protocol (Phase F1). Close-seam ideal → open-wrap cliff (seed 20260719). Mean prolonged R . Primary = ReelSynth export; secondary = AKWF CC0 files.

Method	Export ($n=25$)	AKWF OA ($n=24$)
endpoint-pin (mean)	0.939	0.978
seam_fir3	0.971	0.975
poly seam ($d=3$)	0.967	0.970
no-bake	0.967	0.969
neural favorite	0.911	0.970
DualCosine	0.883	0.936
N2N corrupt→corrupt	0.869	0.974
seq CNN1D	0.839	0.962
seq LSTM	0.769	0.948

5.9 Ablations and compute

RQ: how do search branch-best freezes compare to short isolated GA/PPO budgets, and what compute was used? Table 10 reports two views. Search *branch-best freezes* at the 5k gate come from one hybrid history. Isolated short-budget re-runs (GA-only, PPO-only, GA+PPO, full) use 150 iters on the same search seed with plateau adapt off. Short-budget champions sit below the multi-hour freeze by con-

Table 9: Classical VA seam repairs on the residual protocol (seed 20260719). Canonical holdout: R , SNR/SDR, wrap-jump, edge RMSE. Cliff top-10%: R ($n=410$). Multi: mean R over 20 generative waveforms.

Method	R	SNR	SDR	jump	edge	top10	R multi
BLAMP	0.9718	32.6	32.7	0.873	0.080	0.9667	0.965
no-bake	0.9718	32.6	32.7	0.873	0.080	0.9667	0.965
seam_fir3	0.9610	29.6	29.6	0.438	0.112	0.9429	0.955
PolyBLEP	0.9288	25.9	25.8	0.102	0.205	0.8641	0.924
BLIT/BLEP	0.8680	20.4	20.3	0.004	0.365	0.7116	0.860
DualCosine	0.8249	18.0	17.8	0.292	0.506	0.6574	0.814

Latency (canonical, CUDA): PolyBLEP $\approx$ 1.43 ms/batch; DualCosine $\approx$ 1.06; BLIT/BLEP $\approx$ 123; BLAMP $\approx$ 54. Frozen in `va_seam_blep.json`.

Table 10: Ablations. Left block: search branch-best freeze (run `gpu-rl-arch-20260719T083019Z`, final iter $\approx$ 6,922). Right block: isolated short-budget re-runs (150 iters, seed 1902771841, plateau adapt off). R is prolonged residual on the runner geometry.

Config	Search freeze R	Isolated 150-it R
Full hybrid	0.99093	0.98113
GA-only	0.99016	0.98062
PBT (branch best / n/a isolated)	0.99012	—
PPO-only	0.98924	0.97932
GA+PPO	—	0.98007
combo (branch best)	0.98854	—
NAS (branch best)	0.98677	—
DualCosine (runner)	0.81658	0.81658

Table 11: Compute budget for the reported 5k-gate search freeze (RTX 3090, PyTorch 2.6+cu124, search seed 1902771841). Arch evaluations $\approx$ clean iterations.

Item	Value
GPU	NVIDIA GeForce RTX 3090
Elapsed wall time	$\approx$ 7.92 h
Clean iterations / arch evals	$\approx$ 6,922
Population size	12
Final champion R	0.99093
DualCosine baseline (runner)	0.81658
Peak VRAM (nvidia-smi replay)	$\approx$ 3.29 GiB

Peak memory: max `nvidia-smi memory.used` during a 30-iter full-branch search probe ($\approx$ 3.29 GiB) and champion forward+FitCell replay ($\approx$ 3.17 GiB). The multi-hour search process itself left no memory log. Idle GPU occupancy was already $\approx$ 3.0 GiB from other sessions.

struction. They are labeled as such. Table 11 summarizes the search-campaign compute budget. Peak VRAM is from an `nvidia-smi` poll during a short full-branch search replay (30 iters) plus a champion forward+FitCell replay. The campaign `history.jsonl` did not log memory.

5.10 Meta-learning approach comparison (5k matched budget)

RQ: under a matched 5,000-evaluation budget with LSTM and xLSTM in the searchable bake vocabulary, how do Random

Table 12: Matched 5k-budget outer-loop comparison (search seed 1902771841; LSTM+xDLSTM in bake vocab; reward-mode + HP co-tune in Ours). Champion R vs ideal sibling; ΔR vs DualCosine is one classical gap only. Wall-h: per-method CUDA wall time. LSTM?/xDLSTM?: whether the champion graph included that block.

Method	Champ $R@5k$	ΔR vs DualCosine	Wall-h	LSTM?	xDLSTM?
Random NAS	0.97816	+0.16157	5.72	no	no
Cont. CMA-ES	0.98468	+0.16810	2.28	no	no
Arch REINFORCE	0.98247	+0.16588	25.93	no	no
Aging evolution	0.99079	+0.17421	15.02	no	no
TPE Bayes NAS	0.98106	+0.16448	11.18	yes	yes
Ours (hybrid GA-PPO)	0.99146	+0.17488	8.46	no	no

NAS, Cont. CMA-ES, Arch REINFORCE, Aging evolution, and TPE Bayes NAS compare to **Ours (hybrid GA-PPO)**? Table 12 reports champion prolonged R , ΔR vs DualCosine, wall hours, and whether the champion graph used an LSTM or xLSTM block. At the completed gate, **Ours** leads on absolute champion R ($R \approx 0.99146$, $\Delta R \approx +0.17488$ vs DualCosine ≈ 0.81658 ; $\approx$ 8.46 h), with Aging evolution second ($R \approx 0.99079$). Neither Ours nor Aging evolution selected an LSTM block in the champion graph; Arch REINFORCE / Aging evolution champions used xLSTM, and TPE used LSTM (vocab presence, not a full-stack claim). Figure 6 is the at-a-glance bar comparison; Figure 7 shows champion- R learning curves (x-axis capped at 5,000). Figure 8 shows a holdout wrap-seam heal (seed 20260719) for the refit Ours champion against no-bake, DualCosine, and ideal sibling r^* . Source: `meta_approach_compare.json` (search seed 1902771841).

5.11 Rust sound_bench transfer (20 tiles)

RQ: does the Python-trained favorite transfer to Rust `sound_bench` tiles against the classical non-AI board? Table 13 scores no-bake, `seam_fir3`, DualCosine, other classical rows, and the Python-trained neural favorite on 20 Rust-exported tiles (10 families $\times$ 2 seeds). This closes the prior gap where multi-family tables used only Python generative stand-ins. Honest transfer note: the favorite was searched on Python `make_batch` sine+cliff geometry. On Rust families it still beats DualCosine (mean $\Delta R + 0.063$, Wilcoxon

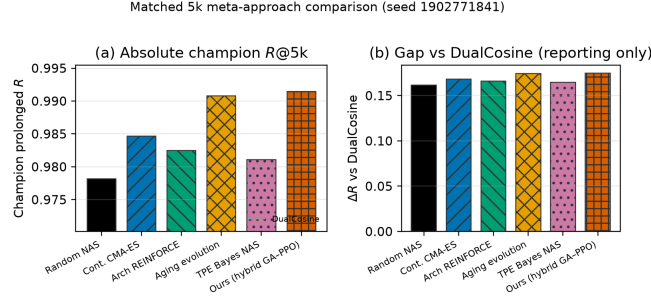


Figure 6: Matched 5k-budget meta-approach comparison (search seed 1902771841): (a) champion prolonged R ; (b) ΔR vs DualCosine (reporting gap only—objective is maximize absolute R toward the ideal sibling). Manuscript names; Okabe–Ito colors with distinct hatches for grayscale print.

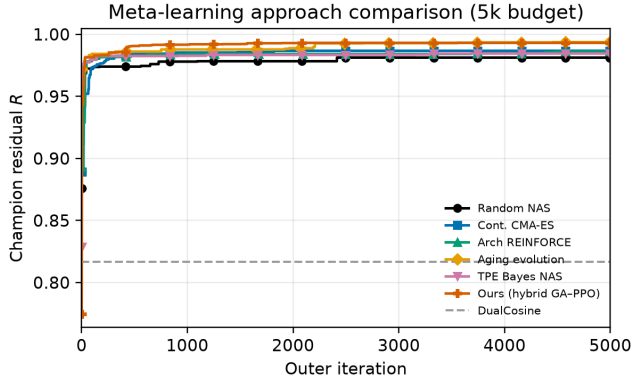


Figure 7: Matched 5k-budget meta-learning comparison: champion residual R versus outer iteration (x-axis capped at 5,000). DualCosine dashed as one classical comparator only—objective is maximize absolute R toward the ideal sibling; rank vs the classical board in Table 4. Accessibility: Okabe–Ito colors with distinct markers for grayscale print.

Table 13: Secondary transfer on 20 Rust `sound_bench` tiles (byte-aligned export). Mean $\pm$ std prolonged R vs the same ideal sibling. Rank by absolute R against classical non-AI rows; ΔR column is vs DualCosine only. Favorite beats DualCosine (mean $\Delta R + 0.063$, $p \approx 0.009$) but trails no-bake / FIR.

Method	R (20 Rust tiles)	ΔR vs DualCosine
no-bake	0.928 ± 0.084	+0.166
seam_fir3	0.896 ± 0.077	+0.134
neural favorite	0.825 ± 0.121	+0.063
classic quadratic	0.788 ± 0.110	+0.026
DualCosine / cosine fade	0.762 ± 0.118	0

$p \approx 0.009$) but trails no-bake and `seam_fir3` on absolute R . The primary SOTA claim remains the Python multi-family matrix.

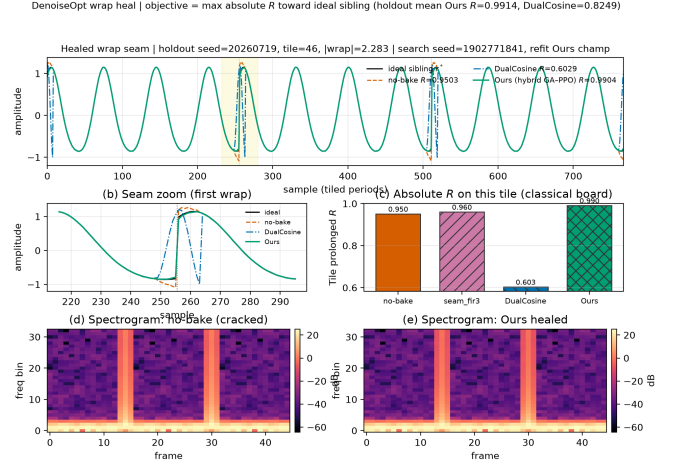


Figure 8: Healed wrap-seam example for the matched-suite **Ours** (hybrid GA-PPO) champion (arch+HP from `hybrid_lstm` summary; FitCell refit with search seed 1902771841). Holdout tile (seed 20260719): prolonged waveforms vs ideal sibling, seam zoom, per-tile absolute R vs no-bake / FIR / DualCosine, and spectrograms before/after heal. Cell weights are refit for visualization (suite checkpoints store arch+HP, not fitted `state_dict`); holdout mean R accompanies the tile view. DualCosine ΔR is reporting only.

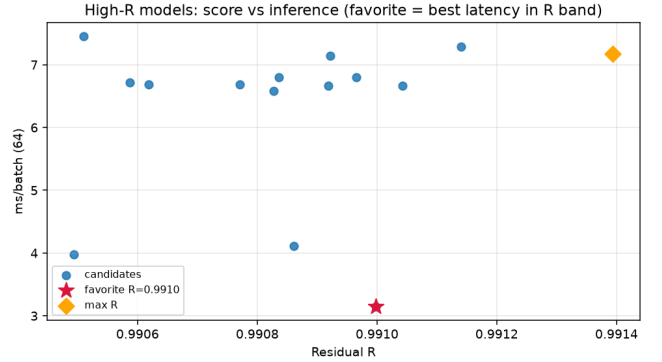


Figure 9: High- R fitted cells on CUDA (batch 64): prolonged residual R versus forward latency (ms/batch). Star marks the favorite (fastest inside the near-max- R band).

5.12 Vibrato playback and spectrograms

RQ: do wrap-seam differences remain visible under *dynamic* pitch (slow vibrato), not only static tiled periods? Figure 18 renders the same holdout cracked period, DualCosine bake, and refit **Ours** (hybrid GA-PPO) champion under shared sinusoidal vibrato ($\pm 3\%$ depth at 5 Hz around 220 Hz). Panels (a–c) show magnitude spectrograms; (d–e) show DualCosine–no-bake and Ours–no-bake differences; (g) reports cycle-local prolonged R alongside a modulated-playback residual under the same pitch envelope. On the annotated high-wrap tile, Ours retains high cycle R and high modulated R , while DualCosine remains the weakest classical comparator. This is objective spectrogram / resid-

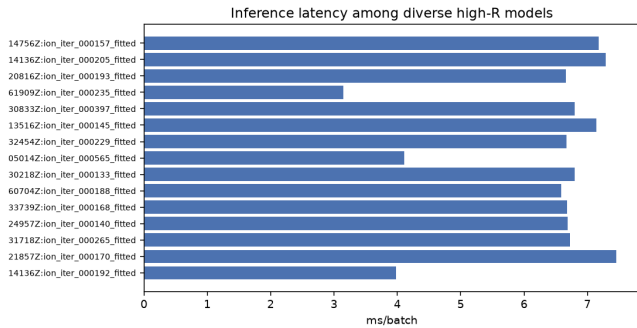


Figure 10: Inference latency (ms/batch, CUDA, batch 64) for diverse high- R fitted models from the inference bench.

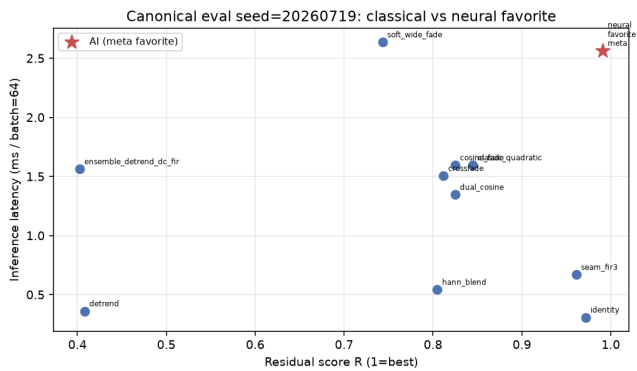


Figure 11: Canonical holdout (seed 20260719, batch 64): classical bake methods and neural favorite in the R vs CUDA ms/batch plane. Star: favorite.

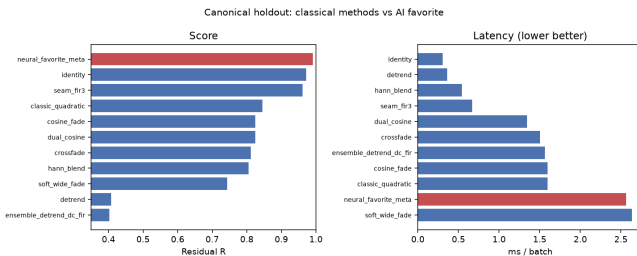


Figure 12: Canonical holdout ranking on prolonged residual R (left) and CUDA latency (right). DualCosine $R \approx 0.825$. Favorite highlighted.

ual evidence only—not a formal listening study. Source: `reelsynth/scripts/bench_vibrato_spectrogram.py`, artifacts under `brand/artifacts/vibrato_spectrogram/`.

5.13 Downloadable hear samples

As an audible companion to Figure 8, we release five fixed-pitch (A4, 3 s, 44.1 kHz mono) wavetable clips for no-bake, DualCosine, and Ours-healed periods from the matched-suite holdout (`reelsynth/brand/artifacts/meta_approach_compound`), rebuild via `scripts/export_meta_hear_samples.py`. Figure 19 shows short waveform excerpts from those WAVs

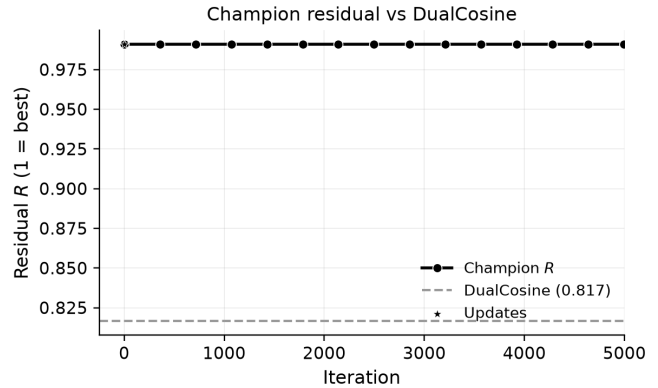


Figure 13: 5k-gate search freeze: champion residual R versus clean search iteration (x-axis capped at 5,000), with DualCosine baseline on the runner geometry (RTX 3090). Accessibility: black line with circle markers; DualCosine as a dashed gray baseline.

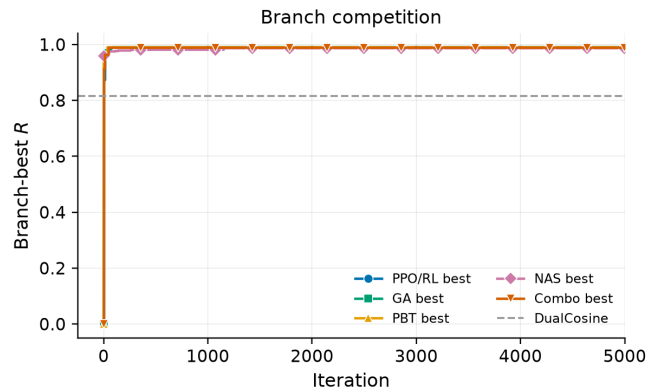


Figure 14: 5k-gate search freeze: per-branch best residual trajectories (PPO/RL, GA, PBT, NAS, combo) versus iteration (x-axis capped at 5,000). Accessibility: colorblind-safe colors plus distinct markers (circle, square, triangle, diamond, inverted triangle) for grayscale print.

with per-clip absolute R . Readers can download and listen; we do *not* report formal MOS/MUSHRA or claimed A/B listening scores.

5.14 Compact wavetable diversity gallery

Beyond the synthetic sine+cliff narrative, Figure 20 shows a compact gallery of true ReelSynth-exported factory periods (one mid-morph frame per bank) and Adventure Kid Waveforms (AKWF) CC0 single-cycle files already used by the wrap protocol. Scored means remain in Table 8 / `real_wt_matrix.json`; this panel is diversity evidence only. No new NAS was run for the gallery.

Hyperparameter sensitivity (Q3). RQ: how sensitive is hybrid champion R to $\pm 50\%$ shifts of Table 2 knobs? We run a **one-at-a-time sensitivity probe** (not a full 5k re-search): 500 outer iterations per config, seed 1902771841, sine+cliff

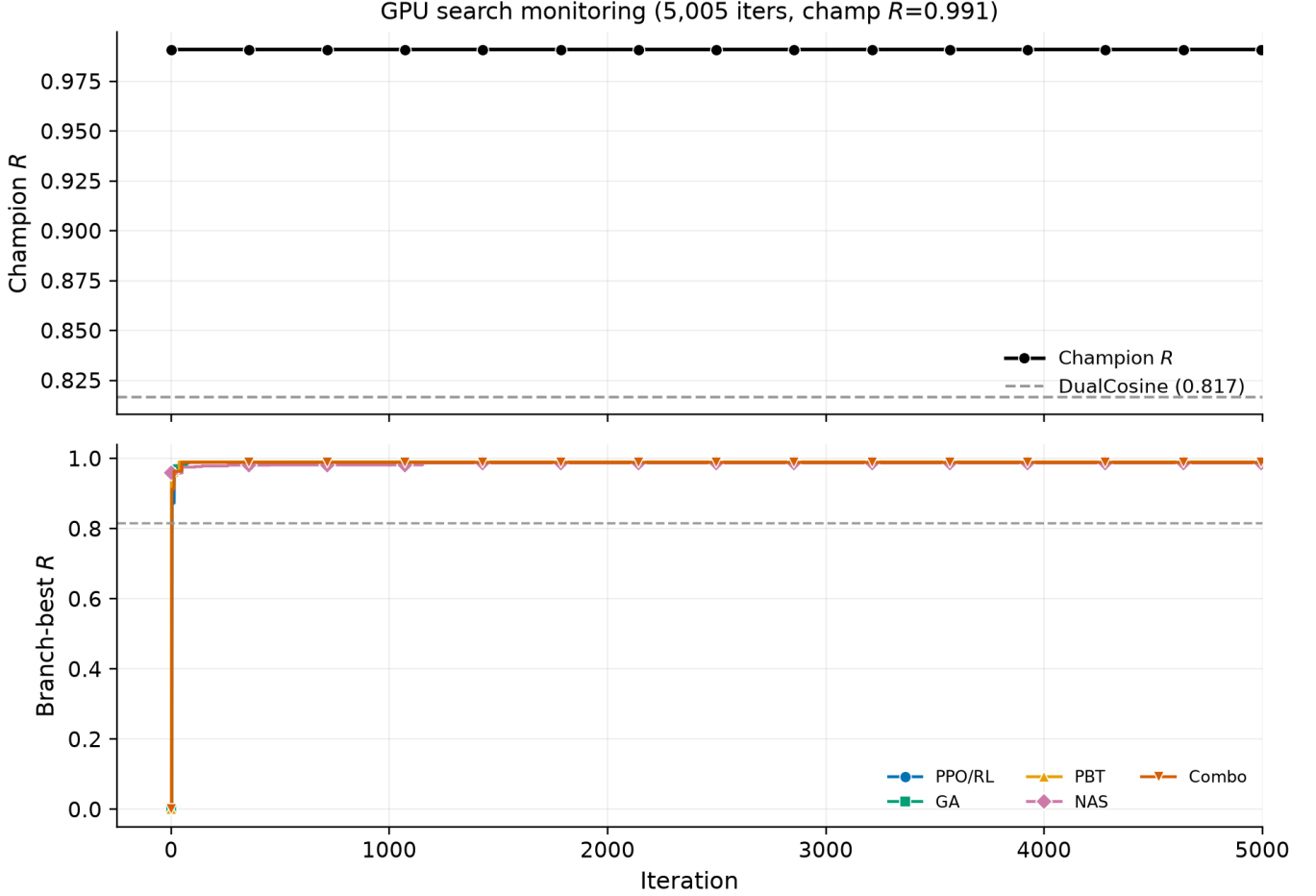


Figure 15: Full-width search monitoring panel at the 5k-gate freeze (first 5,000 iterations): champion R (top) and branch-best trajectories (bottom). Markers and Okabe-Ito colors match Figure 14. Source: `history.jsonl`.



Figure 16: Per-trial residual scatter by search branch for the first 5,000 iterations at the 5k-gate freeze, with champion overlay. Branch markers match Figure 14.

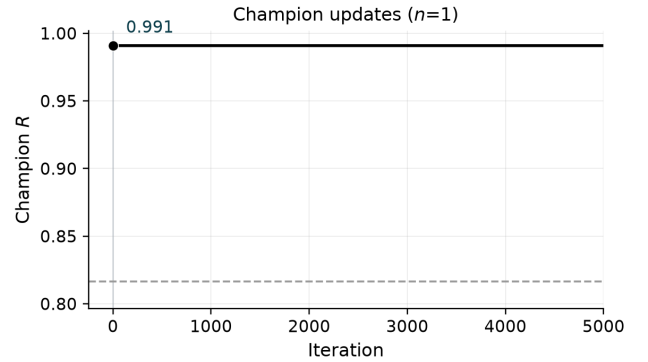


Figure 17: Champion update timeline for the first 5,000 iterations at the 5k-gate freeze. Sparse R labels mark first, mid-peak, and latest improvements for twocolumn legibility.

FitCell matched to the meta-compare protocol (batch 48, fit steps 24). Perturbed knobs: population size n , PPO clip ϵ , fit Adam learning rate, GA second-mutate rate, and PPO entropy coefficient. Table 14 and Figure 21 report champion abso-

lute R versus the Table-2 default (default champ $R \approx 0.9888$; largest $|\Delta R|$ among completed arms ≈ 0.0123 on `lr_m50`). All 11/11 arms completed under this budget.

DenoiseOpt under slow vibrato playback | holdout seed=20260719, tile=46, |wrap|=2.283 | search seed=1902771841

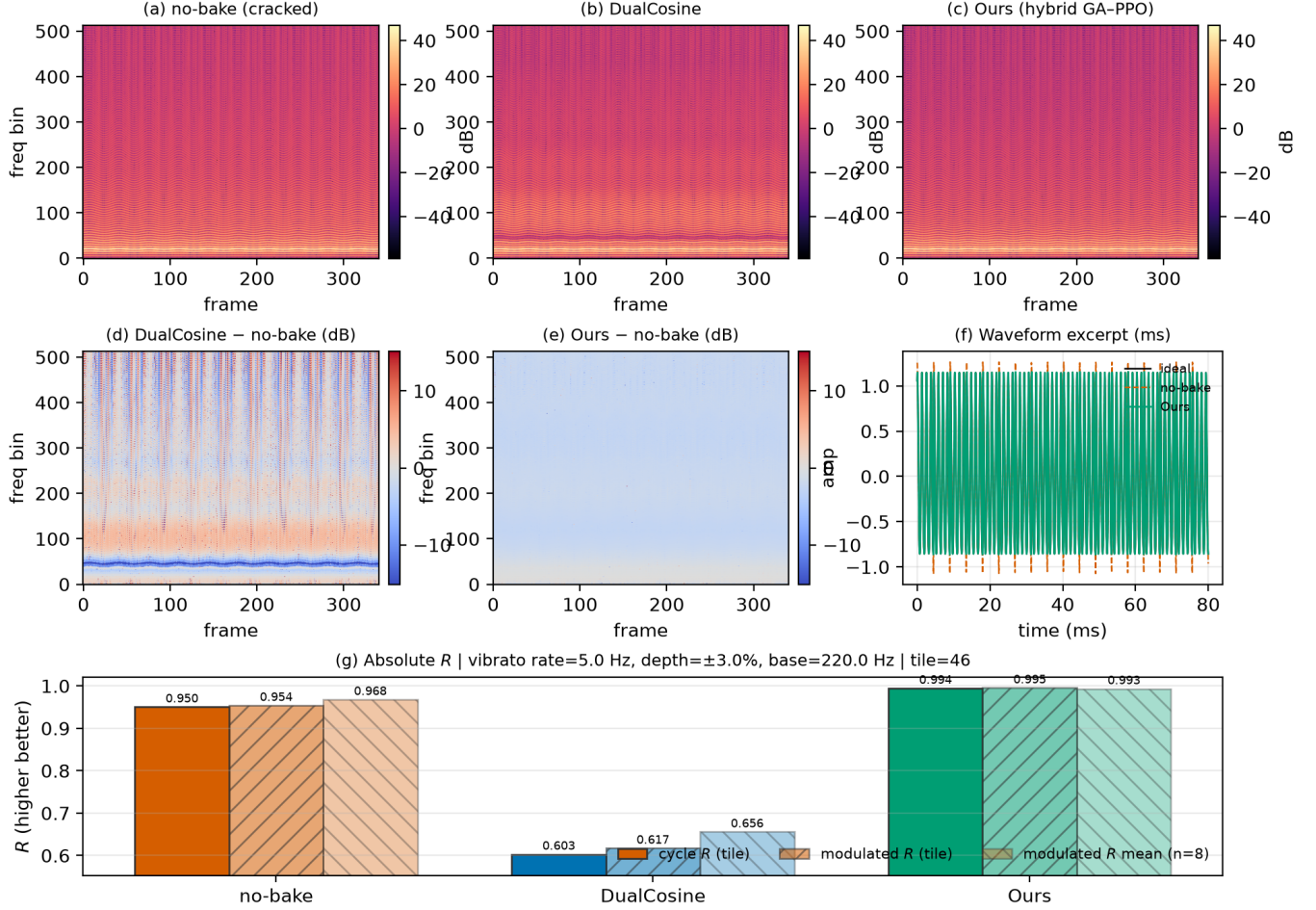


Figure 18: Slow-vibrato spectrogram eval on a high-wrap holdout tile (seed 20260719). (a–c) Magnitude spectrograms for no-bake, DualCosine, and Ours. (d–e) Spectrogram differences vs no-bake. (f) Short waveform excerpt. (g) Absolute R : cycle-local prolonged residual, modulated-playback residual on the same tile, and mean modulated R over top-wrap tiles. Accessibility: Okabe–Ito hues; difference maps use diverging coolwarm. Formal human A/B scores are out of scope.

This is local robustness evidence under the stated budget and seed; the 5k matched-budget ranking (Table 12) remains the primary approach comparison. We do *not* claim that Table-2 defaults are globally optimal—only that $\pm 50\%$ OAT shifts leave champion R within a small band relative to the DualCosine gap (≈ 0.17 on the 5k gate).

6 Discussion

Where residual mass remains. Across the 100,000-cycle procedural bench and lit-combo family stress tests, wrap error is not uniform. The same generator families repeatedly dominate the residual budget: `nonlinear`, `combo`, `triple_mix`, and `extreme_overlay` show the weakest denoise/residual scores, while `harmonic_fft` and `am_fm` are comparatively easy. `open_wrap_bias` can look strong on wrap-energy proxies yet still lag on prolonged residual.

Cliff reduction and ideal-matched tiling are related objectives with different rankings.

No-bake R and hard-cliff honesty. Canonical no-bake $R \approx 0.97$ looks “easy” only if one treats R as a perceptual wrap score or confuses no-bake with the ideal sibling. No-bake is the unrepaired engine scored against r^* ; the ideal withholds the cliff and is never a method output. Under (4), small residual RMS relative to ideal RMS yields high R even when $|x_0 - x_{L-1}|$ is large. That same near-ceiling regime is why tiny ΔR still matters operationally: climbing $0.97 \rightarrow 0.991$ is most of the remaining residual budget, yet raw score differences are small enough that reward shaping and outer-loop hyperparameters must be tuned with the architecture (Section 3.7). Hard-cliff strata make the operative regime explicit: DualCosine drops to $R \approx 0.657$ on the top-10% wrap-jump tiles while the

Audible wrap-seam demos (fixed A4 playback) — downloadable WAVs under `reelsynth/brand/artifacts/meta_approach_compare/hear_samples/`

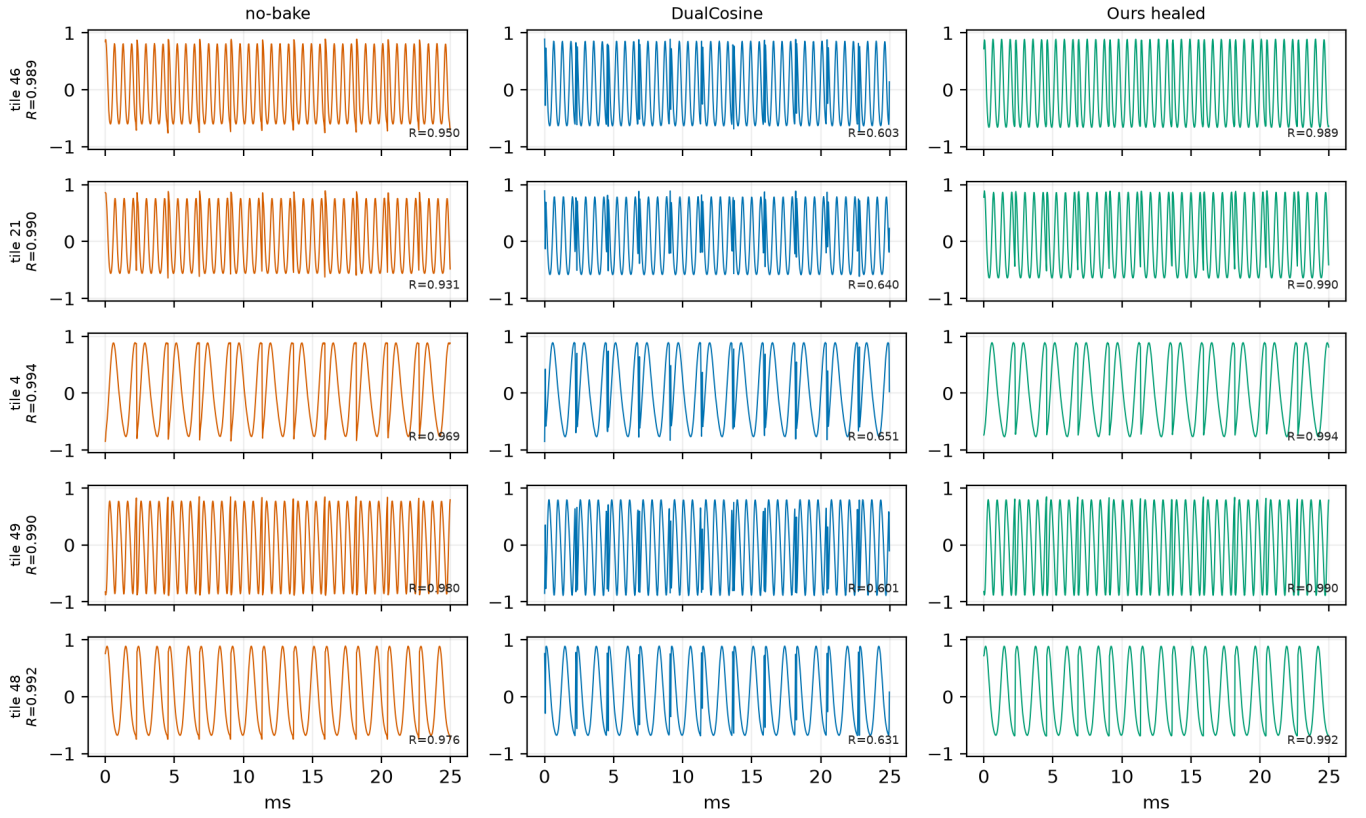


Figure 19: Hear-sample panel from downloadable WAVs under `brand/artifacts/meta_approach_compare/hear_samples/` (five holdout tiles; no-bake / DualCosine / Ours). Waveforms show the first ≈ 25 ms of each clip; R annotations are cycle-local prolonged residuals vs the ideal sibling. Not a formal listening study.

Table 14: One-at-a-time $\pm 50\%$ hyperparameter sensitivity probe (500 outer iterations, seed 1902771841, sine+cliff FitCell). Champion absolute R vs Table 2 default. **Sensitivity evidence only** — not a full 5k re-search per HP.

Config	Champ R	ΔR vs default	Iters
default	0.98882	+0.00000	500
pop_size_m50	0.98136	-0.00746	500
pop_size_p50	0.98073	-0.00809	500
ppo_clip_m50	0.98159	-0.00723	500
ppo_clip_p50	0.99041	+0.00159	500
lr_m50	0.97654	-0.01228	500
lr_p50	0.98138	-0.00744	500
ga_mut2_m50	0.98334	-0.00548	500
ga_mut2_p50	0.99002	+0.00120	500
entropy_m50	0.98912	+0.00030	500
entropy_p50	0.99116	+0.00234	500

favorite stays near 0.9915. Favorite wrap-jump barely moves toward zero on that subset. We claim tiled residual repair toward r^* , not endpoint closure.

N2N stress test and transfer conditions. On the sine+cliff holdout, corrupt $\rightarrow$ corrupt N2N matches the meta favorite within $\approx 10^{-3}$. Sibling-supervised N2N and the LSTM ceiling sit slightly above. That is expected for oracle-label training on the same generator and does not invalidate the hybrid search story: DenoiseOpt discovers compact bake graphs without engine $\rightarrow$ ideal supervision at search time, and the hard-cliff DualCosine collapse contrast remains. Win conditions (`transfer_failures.json`): favorite beats DualCosine on sine cliffs and most AKWF / Rust tiles; no-bake/FIR lead on exported factory geometry (domain shift, not buried). On true ReelSynth exports the searched favorite trails no-bake and FIR; we report that gap rather than claim universal factory superiority.

Wrap-jump vs edge RMSE. Primary claim remains prolonged R . Wrap-jump is diagnostic only. Edge RMSE is the locked seam-local secondary: favorite reduces it on hard cliffs even when wrap-jump stays flat. Endpoint-pin zeros wrap-jump while hurting R , confirming the objectives differ.

Implication for this paper. Mean R (and mean DualCosine gaps) therefore compress a structured difficulty map. Report-

Compact wavetable diversity gallery (existing exports only — no new NAS)

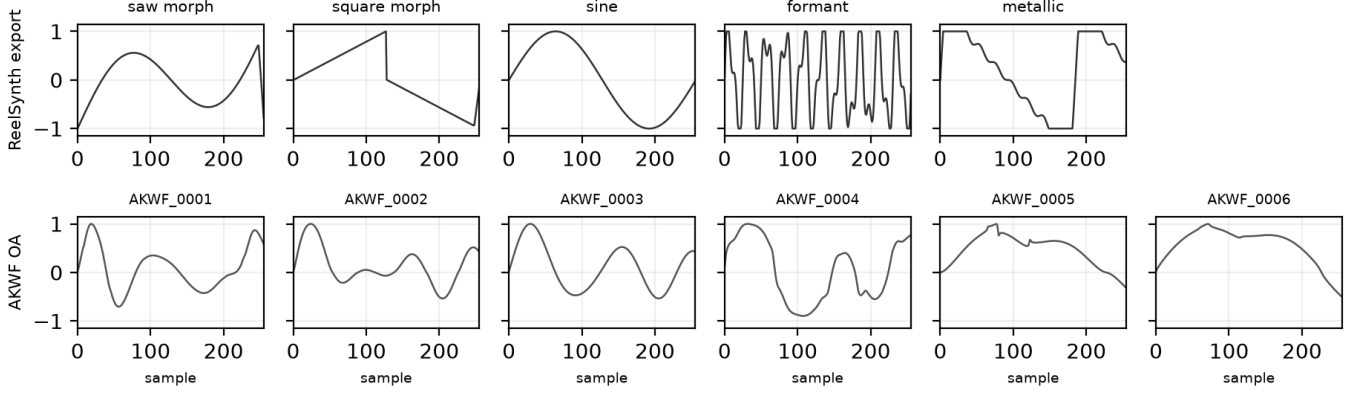


Figure 20: Compact wavetable diversity gallery from existing exports (ReelSynth factory banks, top; AKWF CC0 cycles, bottom). Mid-morph frames per bank; no new meta-search. Quantitative wrap scores: Table 8.

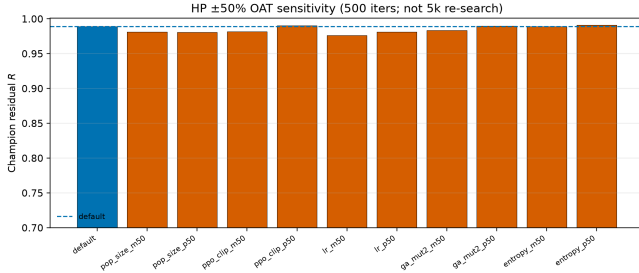


Figure 21: One-at-a-time $\pm 50\%$ hyperparameter sensitivity of hybrid champion R at a 500-iteration budget (seed 1902771841). Blue bar / dashed line: Table 2 default. This is a **sensitivity probe**, not a full 5k re-search per HP.

ing only the search-campaign mean on synthetic sine cliffs understates how much of the remaining $\sim 1\%$ sits in a few hard families, and how DualCosine fails when wrap-jump is large. Hard-cliff strata and wavetable-native transfer (true exports + AKWF) are first-class empirical claims of the present work. The contribution is aimed at DSP / synthesis artifact repair (DAFx / AES / arXiv cs.SD), not general speech-enhancement leaderboards.

Hybrid search vs. plateau. Search traces at the 5k gate show early high champions then long plateaus near $R \approx 0.99$. That plateau is expected under near-ceiling R : further gains are small in absolute units and sensitive to reward shaping, entropy/clip, mutation rates, and plateau-triggered branch reweighting. Plateau adaptation (deeper cells, denser MoE graphs, higher GA/PBT weight) is the operational response inside one campaign. It does not by itself rebalance family-conditional risk. A natural next paper is family-conditional and cliff-conditional meta-learning: train or select denoisers per family, or condition the outer loop on measured wrap statistics, and publish per-family residual curves beside the

scalar champion—together with ablations of reward shaping under the same compressed- R regime.

Hybrid branches. At the reported 5k-gate search freeze, GA and PBT lead branch-best residuals, with PPO close behind and NAS trailing on this seed (Table 10). These per-branch readings are snapshots from a single hybrid run, not causally isolated single-branch experiments. Isolated short-budget re-runs (150 iterations, same search seed) confirm relative ordering under a fixed compute cap (Table 10). The matched 5k outer-loop comparison (Table 12, Figures 6–7) places Random NAS, Cont. CMA-ES, Arch REINFORCE, Aging evolution, and TPE Bayes NAS beside **Ours (hybrid GA-PPO)** under a shared seed and bake vocabulary that includes searchable LSTM/xLSTM blocks. Ours wins the absolute- R gate ($R \approx 0.99146$) with Aging evolution close behind ($R \approx 0.99079$); Cont. CMA-ES is the wall-time leader (≈ 2.3 h) at lower champion R . Figure 8 shows the wrap-seam heal qualitatively on holdout geometry. The multi-family SOTA matrix (Table 5) shows the meta favorite ahead of DualCosine on R , SNR, and SDR across twenty Python generative draws, and ahead of no-bake / FIR on absolute R on that same generator. Fixed MLP/CNN-on- R baselines beat DualCosine and trail the searched favorite. On Rust `sound_bench` tiles (Table 13) the same favorite beats DualCosine (mean $\Delta R = +0.063$) and trails no-bake / FIR. That transfer gap matches the narrow Python training geometry.

Review narrative (v8). Peer review asked for Methods clarity, formal ideal-sibling math, broader-than-static eval, and HP justification for meta-search. v8 answers with slim IM-RaD Methods plus Algorithms 1–9 in the appendix (Section 3, Appendix A); formal Θ and $G(\text{seed}, \text{cliff})$ siblings (Q1–Q2); vibrato spectrograms, downloadable hear WAVs, and a compact real-WT gallery (Sections 5.12–5.14); and a 500-iteration one-at-a-time $\pm 50\%$ HP sensitivity probe (paragraph 5.14) that measures local robustness of Table 2 defaults under seed

1902771841. That probe is *not* a second 5k re-search and does not re-rank approaches. Sci/eng wrap transfer remains a classical-board pilot (Appendix B), not a deep-SOTA CWRU/ECG claim. Audible demos support informal inspection only—no formal human listening study.

7 Tooling and AI use

This section discloses research and writing tools, including AI systems, with scope and intent.

Literature discovery. Open literature was gathered with the Klaut Research Gateway (local OpenAlex / Crossref / arXiv providers) under queries spanning virtual-analog anti-aliasing, unsupervised restoration, HPO/PBT, RL-GA hybrids for NAS, and audio denoise architectures. Harvest JSONs live in the companion meta repository. Source PDFs were downloaded for every bibliography entry (artifacts/literature_oa/pdfs/). Paywalled-only items were dropped or replaced.

Paper drafting. Section planning, subsection drafts, and revision passes used Klaut `research_paper_*` tools with local Ollama (qwen3.5:9b) when available, under an explicit scientific workflow (plan → write → revise → export) into an arXiv-style twocolumn template. AI drafts were treated as scaffolding. Claims, numbers, and citations were checked against harvest records, PDF analyses, and measured artifacts. Final prose was edited for DSP terminology and to remove generic filler.

Why AI was used. (1) Scale literature triage across DSP and AutoML without pretending every high-cite hit is in-domain. (2) Accelerate IMRaD scaffolding so human effort focuses on metric honesty (residual vs DualCosine, family hardness, gate vs final 5k-gate search claims). (3) Keep an auditable tool chain (plans, PDF snippet analyses, citation flags) over opaque chat-only writing.

Where AI was *not* used. Model fitting, residual scoring, DualCosine baselines, lit-combo / hybrid GPU search, and inference-time benchmarks are conventional code (Rust / PyTorch) with deterministic logs and JSON artifacts. AI did not invent experimental numbers.

Other tools. Rust/cargo for the procedural bench and meta binaries. PyTorch CUDA for hybrid GA-PPO search. matplotlib for figures. MiKTeX pdflatex for PDF builds. GitHub for versioned papers and artifacts.

Hear samples and vibrato figures. Audible wrap demos: `reelsynth/brand/artifacts/meta_approach_compare/hear_samples/` (rebuild: `scripts/export_meta_hear_samples.py`).
 Vibrato spectrogram bench:
`scripts/bench_vibrato_spectrogram.py` →
`brand/artifacts/vibrato_spectrogram/` (mirrored into `paper/v8/figures/`). Listening panels for the manuscript are spectrogram + downloadable WAV citations only.

8 Broader impact and reproducibility

Broader impact. This work targets cycle-local wavetable seam repair for virtual-analog / software-instrument workflows. Misuse potential is low. The method does not enhance speech for surveillance, does not remove watermarks from protected media, and operates only on short synthetic periods. Training and search use procedural sine+cliff generators. No private studio stems are required for the reported tables. The main resource cost is GPU time on a consumer RTX 3090 (order of hours for the 5k-gate freeze, Table 11). Authors should not over-claim speech enhancement or production mastering quality from these metrics.

Reproducibility. Holdout seed 20260719. Hybrid search seed 1902771841. Primary code: <https://github.com/reeldemo/reelsynth> (`scripts/overnight_gpu_rl_arch.py`, `scripts/bench_sota_matrix.py`, `scripts/metrics_snr_sdr.py`, `scripts/bench_va_seam_blep.py`). Paper companion: <https://github.com/reeldemo/denoise-opt-meta>. Dependencies for the CUDA benches: Python 3.12, PyTorch 2.6+CUDA 12.4, NumPy/matplotlib. Rust/cargo builds the separate `sound_bench` hardness probe. JSON artifacts under `brand/artifacts/` (`sota_matrix.json`, `canonical_eval_dataset/`, `search_history.jsonl`, `va_seam_blep.json`) freeze the numbers in Section 5. PESQ/STOI are omitted for non-speech cycles (domain mismatch). MUSHRA was not run.

8.1 Funding

This research received no external funding.

8.2 Author contributions (CRediT)

Julian M. Kleber: Conceptualization, Methodology, Software, Investigation, Formal analysis, Data curation, Writing (original draft), Writing (review and editing), Visualization.

8.3 Data availability

Frozen evaluation artifacts (JSON matrices, search history, fitted champions) are published with the companion repositories: <https://github.com/reeldemo/reelsynth> under `brand/artifacts/`, and <https://github.com/reeldemo/denoise-opt-meta> under `artifacts/` and `paper/v7/figures/`. All reported tables regenerate from those files plus the scripts named above. No private studio recordings were used.

8.4 Competing interests

None declared.

9 Limitations

Scope and evaluation limits. The prolonged residual R is defined against a procedural no-cliff sibling and does not capture perceptual quality directly. Formal listening tests (MOS,

MUSHRA) and claimed A/B listening scores are outside the scope of this work; v8 ships spectrograms and downloadable WAVs for informal audition only. PESQ and STOI are excluded because the primary cycles are non-speech synthetic wavetable tiles; speech-quality metrics on sine+cliff families would be a domain mismatch. MUSHRA was not run (requires human listeners). Any perceptual or speech-proxy secondary study would need separately labeled open-access speech snippets and remains out of scope for this narrow-seam paper.

- Residual uses a procedural ideal (no-cliff sibling). Perceptual MOS and formal listening panels are out of scope.
- Bake metrics leave formal listening tests and speech quality metrics to separate studies.
- **PESQ / STOI / MUSHRA deferred.** Primary cycles are non-speech wavetable tiles. Speech metrics would domain-mismatch sine+cliff families, so PESQ/STOI numbers are absent here (none invented on synthetic tiles). MUSHRA needs human listeners and was *not run*.
- **No LibriSpeech / MUSDB.** Wavetable-native realism uses true ReelSynth-exported factory periods (primary) and external AKWF CC0 single-cycle WAVs (secondary) under the wrap protocol. Procedural stand-ins are tertiary smoke only. Speech/music enhancement corpora remain off by design.
- **HP sensitivity is not a 5k re-search.** Paragraph 5.14 reports one-at-a-time $\pm 50\%$ probes at 500 outer iterations (seed 1902771841). It measures local robustness of Table 2 defaults; it does not re-prove the matched 5k approach ranking.
- **No deep SOTA on CWRU/ECG.** Appendix B is a classical-board wrap-heal pilot only.
- Classical poly seam fitter is reported; lightweight SSM seq models are deferred (timebox); LSTM remains the supervised seq ceiling.
- On exported factory / Rust tiles the meta favorite can trail no-bake or FIR (domain shift). Transfer win-rates are frozen in `transfer_failures.json`; we do not claim universal factory superiority.
- The CUDA search runner scores Python `make_batch` sine+cliff draws. A separate Rust `sound_bench` export of 20 family tiles (byte-aligned generator) is a secondary transfer block. The Python generative SOTA matrix is the primary multi-family table.
- Per-trial architecture width is capped for search throughput. Full Demucs-/diffusion-scale stacks were screened out (domain mismatch vs $L=256$ periodic seams).

- We do not claim wrap-closure or hybrid-search convergence theorems. Empirical gains are protocol-bound. Endpoint-pin shows wrap-jump can be zeroed while R falls.
- The 5k-gate freeze reports measured champion and branch-best residuals. Isolated GA/PPO/GA+PPO/full re-runs use a short labeled budget (same search seed). Those short runs are a different protocol from the multi-hour freeze.
- Classical VA/BLEP lineage cites are OA author or proceedings PDFs (stilson1996, nam2009polyblep, esqueda2016blamp). Cycle-local BLIT/BLEP, PolyBLEP, and BLAMP seam repairs are now scored (`va_seam_blep.json`); they beat DualCosine on prolonged R in places but trail no-bake / FIR / the meta favorite on the value-cliff geometry.

9.1 Outlook: follow-up publication

A self-contained follow-up can treat *which sounds fail* as the primary object of study: stratified residual heatmaps over the ten procedural families, cliff-size conditioning, and family-specialist or mixture-of-experts denoisers selected by the same RL-GA outer loop. That paper would cite the present mean- R hybrid protocol as the baseline and ask whether beating $R > 0.999$ is mainly an architecture problem or a coverage problem over hard families.

10 Conclusion

DenoiseOpt casts wavetable wrap discontinuity as cycle-local seam repair under a prolonged residual R , with a hybrid GA-PPO(+PBT) outer loop over bake cells. At the 5k-gate search freeze the champion reaches $R \approx 0.9909$ versus DualCosine ≈ 0.8166 . On the frozen holdout a compact favorite reaches $R \approx 0.991$ versus DualCosine $R \approx 0.825$; across twenty multi-family draws, mean $\Delta R \approx +0.162$ (Wilcoxon $p \approx 8.9 \times 10^{-5}$). Hard-cliff strata keep favorite $R \approx 0.9915$ while DualCosine falls to ≈ 0.657 . Same-generator N2N / seq baselines bound the supervised ceiling. Wavetable-native transfer covers factory and OA instrument cycles under the wrap protocol. Scope remains wavetable seam restoration for DSP / synthesis venues. Code: <https://github.com/reeldemo/reelsynth>, [https://github.com/reeldemo/denise-opt-meta](https://github.com/reeldemo/denoise-opt-meta).

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Algorithm 1 ResidualScore**Require:** Ideal cycle r^* , baked cycle y , periods N

- 1: $I \leftarrow \text{Tile}(r^*, N)$
- 2: $Y \leftarrow \text{Tile}(y, N)$
- 3: $r_{\text{rms}} \leftarrow \text{RMS}(Y - I)$
- 4: $i_{\text{rms}} \leftarrow \max(\text{RMS}(I), \epsilon)$
- 5: **return** $\text{clamp}(1 - r_{\text{rms}}/i_{\text{rms}}, 0, 1)$

Algorithm 2 DualCosine baseline**Require:** Engine cycle x

- 1: Fade both ends of x with raised-cosine windows of width W
- 2: **return** $\text{ResidualScore}(r^*, x_{\text{faded}}, N)$

Algorithm 3 No-bake (passthrough) control**Require:** Engine cycle x (open-wrap cliff present)

- 1: **return** $\text{ResidualScore}(r^*, x, N)$ $\triangleright$ no operator: $x \mapsto x$

A Algorithms

Companion pseudocode for the residual score, classical controls, FitCell, and hybrid outer-loop branches. Canonical narrative lives in Section 3; these listings mirror docs/PSEUDOCODE.md and the CUDA search runner (overnight_gpu_rl_arch.py / denoise_meta_evo.py).

B Transfer pilot on CWRU bearings and MIT-BIH ECG

We briefly transfer the winning outer loop (**Ours / hybrid GA-PPO**, approach code `hybrid_lstm`) to public cycle-local wrap tasks outside musical wavetables: Case Western Reserve University (CWRU) bearing acceleration periods and MIT-BIH ECG beat windows ($L=256$; prolonged residual R vs an ideal sibling constructed by domain-specific periodization). Table 15 summarizes holdout-refit R against a *classical* board (no-bake, DualCosine, fades, endpoint pin, seam FIR, and domain classical proxies). Artifacts: `reelsynth/brand/artifacts/signal_heal_transfer`, prose source: `denoise-opt-meta/docs/SIGNAL_HEAL_TRANSFER_PILOT.md`.

Classical-board disclaimer (required). This appendix compares against classical / domain-proxy baselines only. It is *not* a claim of beating published deep learning SOTA on CWRU diagnosis or ECG enhancement (e.g., CycleGAN, BeatDiff, or clinical denoisers) unless those models were executed under the same residual protocol—they were not. Outer-loop budget is modest (250 outer iters / dataset, `fit_steps = 40`): a transfer pilot, not a multi-day domain campaign. Real content is z-scored per period; absolute clinical/engineering scale is not preserved.

On CWRU, classical DualCosine / fades *hurt* vs no-bake; Ours only narrowly beats no-bake on holdout—transfer is weak on that wrap construction. On MIT-BIH ECG, DualCosine collapses morphology while Ours leads the classical board (endpoint pin is the strongest classical comparator). Pri-

Algorithm 4 MoE soft gating over bake experts**Require:** Cycle x , expert cells $\{E_k\}_{k=1}^K$, gate logits $g \in \mathbb{R}^K$

- 1: $w \leftarrow \text{softmax}(g)$
- 2: $y \leftarrow \sum_{k=1}^K w_k E_k(x)$
- 3: **return** y

Algorithm 5 FitCell (inner loop)**Require:** Architecture `cfg`, fit steps T , batch size B

- 1: `cell` $\leftarrow$ `SeamCell(cfg)`; `opt` $\leftarrow$ `Adam(cell.params)`
- 2: `prev` $\leftarrow$ `null`; `patience` $\leftarrow$ 0
- 3: **for** $t = 1, \dots, T$ **do**
- 4: `ideal`, `eng` $\leftarrow$ `MakeBatch(B)`
- 5: `out` $\leftarrow$ `ApplyOps(eng, cell, cfg.ops)`
- 6: $R \leftarrow \text{meanResidualScore}(\text{ideal}, \text{out})$
- 7: `loss` $\leftarrow 1 - R$
- 8: **if** `finite(loss)` **then** `AdamStep(opt, loss)`
- 9: **end if**
- 10: **if** `prev` $\neq$ `null` and $|\text{prev} - R| / \max(|\text{prev}|, \epsilon) < 10^{-4}$ **then**
- 11: `patience` $\leftarrow$ `patience` + 1; **if** `patience` ≥ 3 **then**
- 12: **break**
- 13: `patience` $\leftarrow$ 0
- 14: **end if**
- 15: `prev` $\leftarrow R$
- 16: **end for**
- 17: **return** `cell`, R

Algorithm 6 GA tournament step**Require:** Population P , tournament size k , crossover rate p_c , mutation rate p_m

- 1: Sample k genomes; `parents` $\leftarrow$ top-2 by R
- 2: `child` $\leftarrow$ `Crossover(parents)` with prob. p_c , else clone elite
- 3: `Mutate(child)` with rate p_m (ops, depth, cell type, θ)
- 4: Fit and evaluate `child`; insert into P (replace worst if full)
- 5: **return** updated P

mary paper claims remain cycle-local *wavetable* seam restoration.

Algorithm 7 PPO architecture update

Require: Actor-critic π_ϕ , clip ε , experience buffer of mutate actions

- 1: Sample action a editing ops / hyperparameters from π_ϕ
- 2: Fit proposed cfg; observe centered advantage $\rho = R - R_{\text{DualCosine}}$ $\triangleright$ objective remains maximize R
- 3: Advantage $\hat{A}$ from critic; ratio $r_\phi = \pi_\phi(a) / \pi_{\phi_{\text{old}}}(a)$
- 4: Update ϕ with clipped surrogate $\min(r_\phi \hat{A}, \text{clip}(r_\phi, 1 - \varepsilon, 1 + \varepsilon) \hat{A})$
- 5: **return** updated π_ϕ

Algorithm 8 PBT exploit-mutate

Require: Population P , exploit quantile q , mutate scale σ

- 1: Rank P by R ; for each bottom member, copy hyperparameters from a top- q elite
- 2: Perturb copied hyperparameters by multiplicative noise of scale σ
- 3: Refit briefly; write back into P
- 4: **return** updated P

Algorithm 9 HybridMetaSearch

Require: Iteration budget T_{out} , population size n

- 1: $P \leftarrow \text{InitPopulation}(n)$; $\pi \leftarrow \text{ActorCritic}()$; champ $\leftarrow$ null; boredom $\leftarrow 0$
- 2: **for** $it = 1, \dots, T_{\text{out}}$ **do**
- 3: branch $\leftarrow \text{Rotate}(\text{ppo}, \text{ga}, \text{pbt}, \text{nas}, \text{combo})$
- 4: cfg, hp $\leftarrow \text{Propose}(\text{branch}, P, \pi)$
- 5: cell, $R_{\text{fit}} \leftarrow \text{FitCell}(\text{cfg}, \text{hp}, \text{fit_steps})$
- 6: $R_{\text{eval}} \leftarrow \text{EvalCell}(\text{cell})$
- 7: $R \leftarrow \frac{1}{2}R_{\text{fit}} + \frac{1}{2}R_{\text{eval}}$ (+ optional depth/MoE bonus)
- 8: UpdatePopulation(P , cfg, R); PPOUpdate(π , centered $\rho = R - R_{\text{DualCosine}}$)
- 9: **if** $R > \text{champ}.R$ **then** champ $\leftarrow$ (cfg, cell, R); boredom $\leftarrow 0$
- 10: **else** boredom $\leftarrow$ boredom + 1
- 11: **end if**
- 12: **if** boredom ≥ 1000 **then** PlateauAdapt; boredom $\leftarrow 0$
- 13: **end if**
- 14: **end for**
- 15: **return** champ

Table 15: Transfer pilot holdout-refit prolonged R (higher better) on classical boards. Search-time champion R was slightly higher; holdout-refit is the comparison number. Disclaimer: classical board only—not deep SOTA.

Method	CWRU bearings	MIT-BIH ECG
Ours (hybrid_lstm)	0.8705	0.8792
no-bake / bad-COT passthrough	0.8675	0.6403
endpoint-pin (mean)	0.7990	0.7796
spline join (ECG proxy)	—	0.7134
seam_fir3	0.7865	0.6615
DualCosine	0.7846	0.2009
linear fade	0.7929	0.2719
soft Hann periodize	0.7703	0.1479
beat-average SBMM-lite (ECG)	—	0.2135

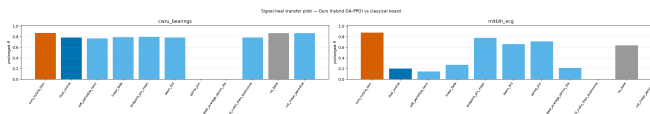


Figure 22: Transfer-pilot classical-board summary (CWRU / MIT-BIH). Bars are holdout-refit prolonged R . Not a deep-SOTA claim; see Appendix B disclaimer.

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